

Plasticity

Plasticity

- Plasticity is an intrinsic property of the human brain and represents an “invention of evolution” that enables the nervous system to overcome the constraints imposed by its genome, allowing it to adapt to environmental pressures, physiological changes, and experience.
- Plasticity is the mechanism underlying the development and learning of cognitive abilities, but it can also contribute to the emergence of pathology.

Plasticity

- It is the ability of neural circuits to modify their structure and function in response to stimuli, both during development and throughout adult life.
- During the early stages of brain development, plasticity is extremely high: some neuronal circuits are selected while others are eliminated.
- In adulthood, many circuits remain largely stable, yet neuronal populations continue to exhibit a certain degree of dynamism, reorganizing under the influence of the external environment to meet specific motor, sensory, cognitive, or emotional demands.

Plasticità dello sviluppo

- **Development of the Embryonic Brain**
 - CNS as “a fluid filled tube”
 - proliferation and synaptogenesis
 - migration
 - differentiation and maturation
 - myelination
 - excess production of neurons
 - development of connectivity
 - NGF and other neurotrophins
 - pruning



25 days



35 days



40 days



50 days



100 days



5 months



6 months



7 months

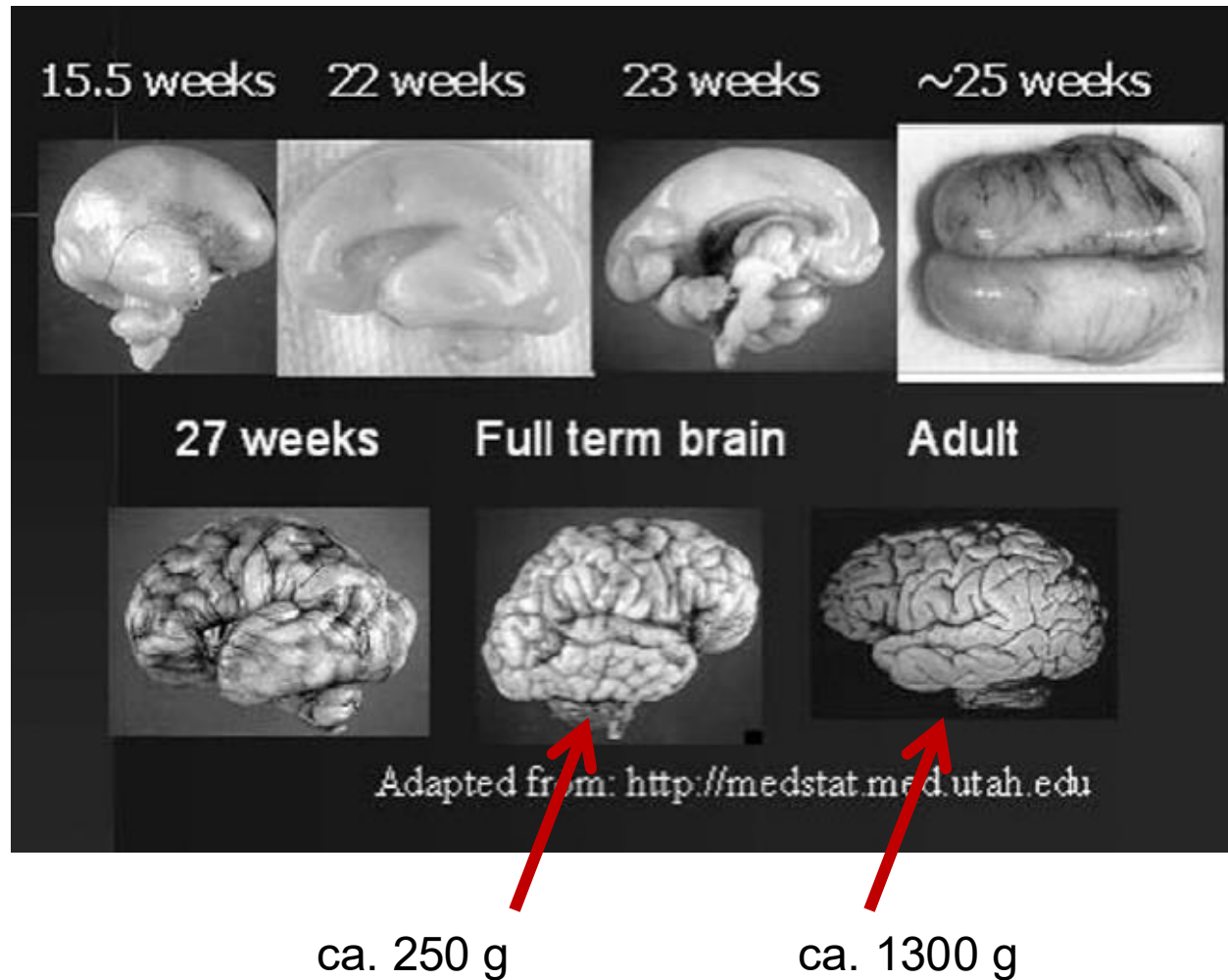


8 months



9 months

Structural and functional development

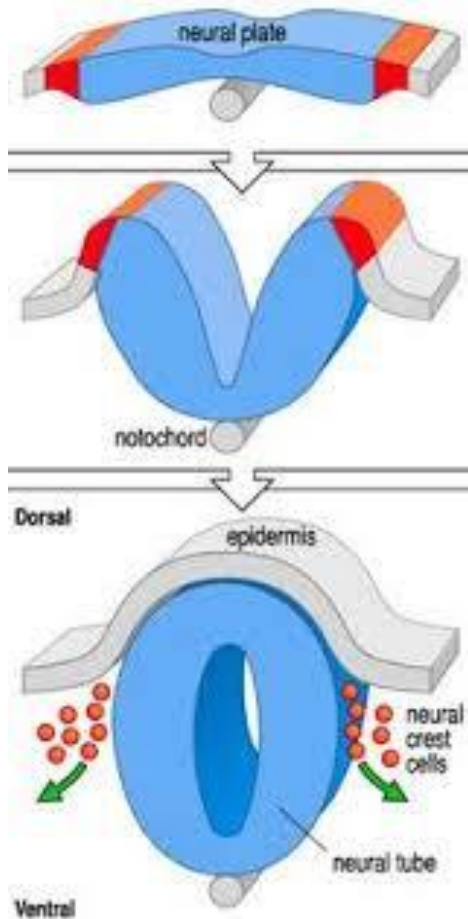


Embryonic development (E13-20)

Following several cell migration steps,
three main layers emerge

- **mesoderm** cells give rise to the structures including muscles, bones, cartilage and the vascular system.
 - **epidermal ectoderm** cells give rise to structures including the skin, nails and sweat glands.
 - **neuroectoderm cells** give rise to the brain and the nervous system.
-

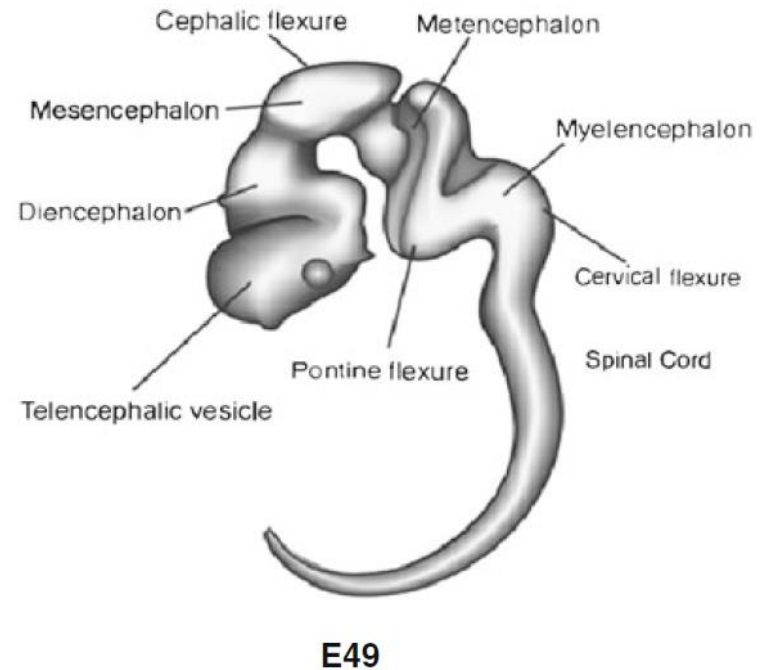
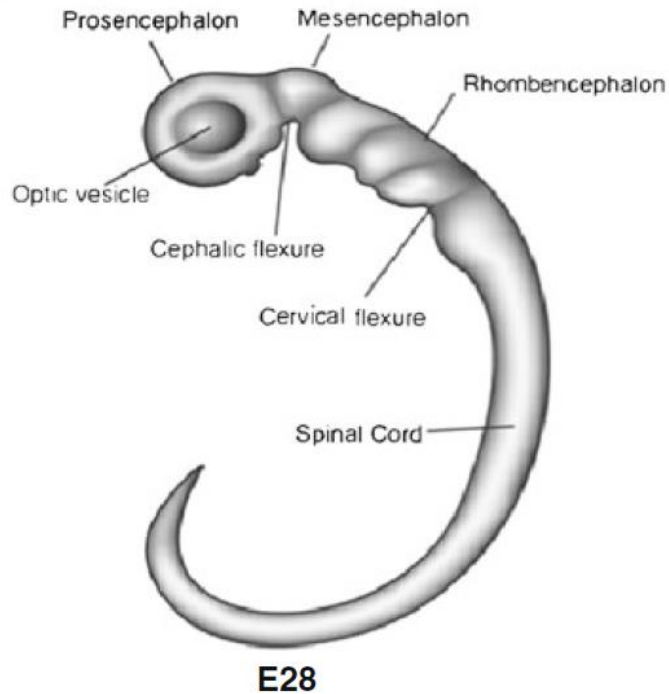
Structural and functional development (E19)



- **neural plate**: part of the three-layer embryo that contains the neural progenitor cells which is located between two ridges
- **neural tube**: the ridges grow, bent towards each other and fuse
- the hollow of the neural tube will later grow and change its shape and will finally turn into the ventricular system. The location of the neural progenitor cells is called the **ventricular zone (VZ)**

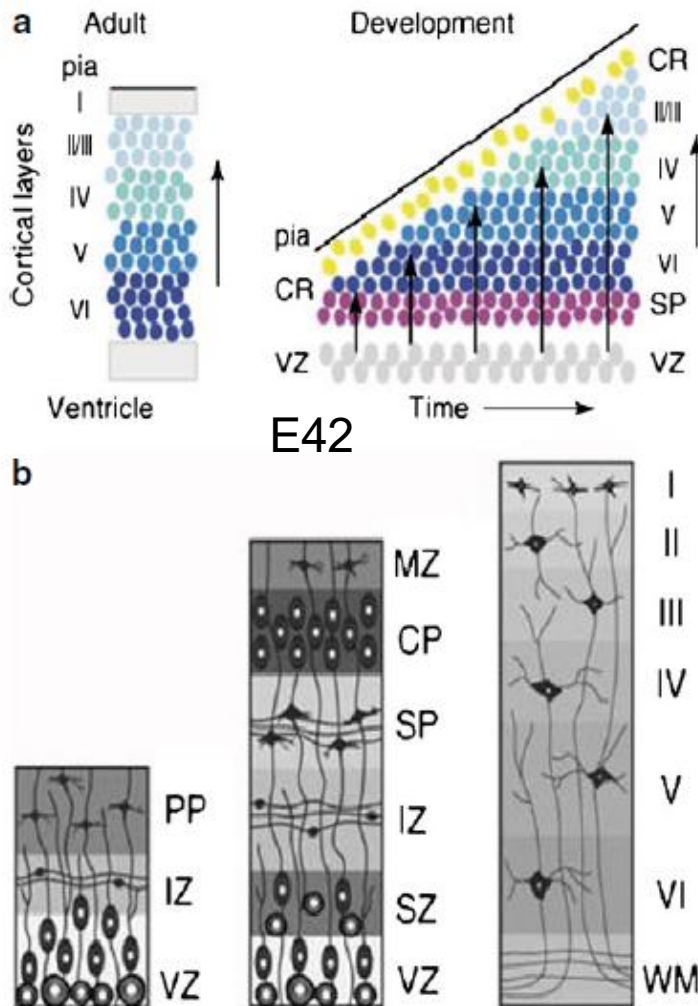
Embryonic development (E28-E49)

E



- After complete closure of the neural tube the embryo begins to expand, particularly in the anterior direction
- the prosencephalon differentiates into the telencephalon and the diencephalon
- the rhombencephalon differentiates into the metencephalon and the myelencephalon

Neural Migration

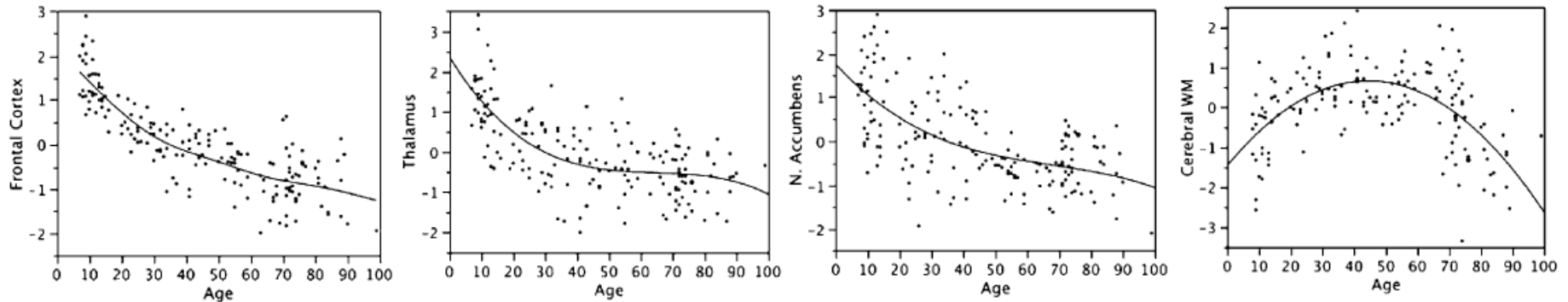


- First neurons that leave the VZ form the **preplate (PP)** that later splits into the **marginal zone (MZ)** and the **subplate (SP)**
- MZ begins to form the **cortical plate (CP)**
- the marginal zone and the subplate finally disappear
- **The brain is built according to the inside-out principle:** new neurons always move on top of existing neurons.
- Finally the **six layer structure** of the cortex emerge. The mature brain has six cortical layers (I-VI), but **none of the embryonic structures** (MZ, SP, VZ). **The intermediate zone (IZ)** has become a mature **white matter layer (WM)**

Regressive events in prenatal brain development

- Neural cell death: 50% of the produced neurons are lost, mostly prenatally (in some regions up to 70%). This process is called **apoptosis = programmed cell death**: cell intrinsic event comprising an ordered sequence of events
 - **Neurotrophic hypothesis**: neurons that establish effective connections are able to obtain more neurotrophins and are more likely to survive
 - establishing effective and functional neural circuits
 - correcting errors in neural production and migration
 - Transient zones such as the MZ and SP as well as a high number of neural progenitor cells die.
-

Imaging studies of brain morphology

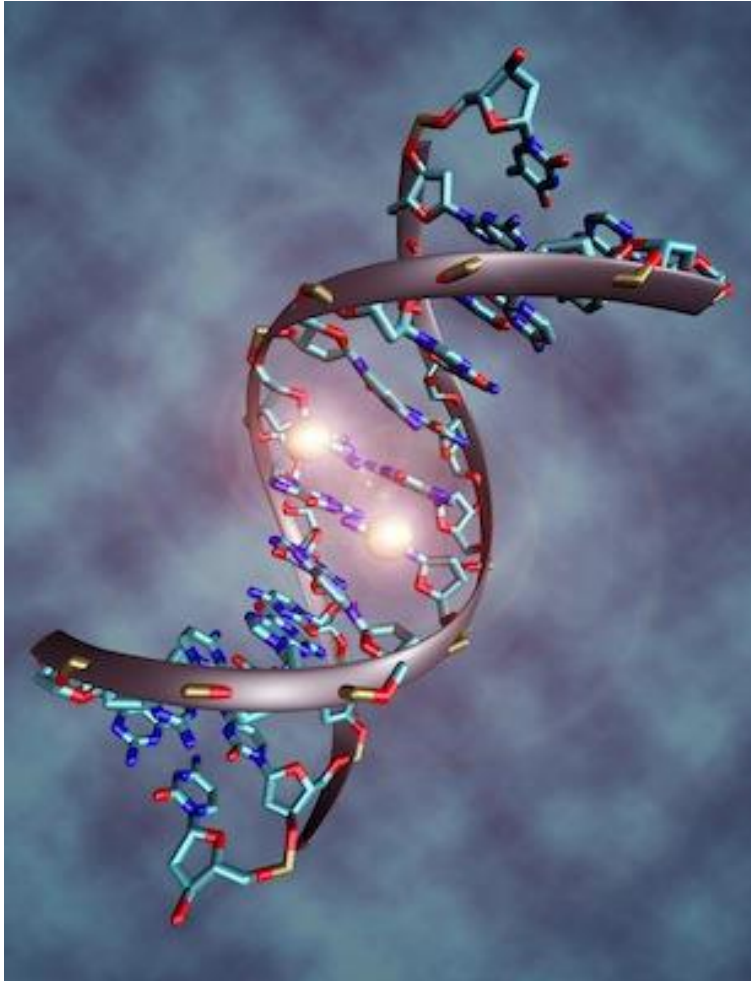


- 1-3 Decrease in volume of frontal cortex, thalamus and nucleus accumbens (cortical thinning)
- 4 Increase in white matter (Increasing fractional anisotropy, FA) from 5 to 12 years in the corpus callosum (splenium) the inferior and superior longitudinal fasciculi, inferior fronto-occipital fasciculi

In General..

- Brains do not typically develop in the absence of **critical genetic signaling** AND they do not typically develop in the absence of **essential environmental input**
 - Development is characterized by **growth, selection and differentiation**; **it is exquisitely dynamic, interdependent, and has probabilistic nature** (Lewkowicz, 2011)
-

Epigenetics



Epigenetics: Mechanisms

Transcription factor: protein that is able to bind to the regulatory region of the gene and thus it activates or represses gene transcription

Epigenetics-> the environment is able to modify pre-existing transcriptional factors through chemical modifications

DNA methylation: addition of a methyl group to the cytosine.

As a result, the **DNA transcription is changed**

How early life experience is written into DNA



Epigenetics

nature
neuroscience

Parental olfactory experience influences behavior and neural structure in subsequent generations

Brian G Dias^{1,2} & Kerry J Ressler¹⁻³

Using olfactory molecular specificity, we examined the inheritance of parental traumatic exposure, a phenomenon that has been frequently observed, but not understood. We subjected F0 mice to odor fear conditioning before conception and found that subsequently conceived F1 and F2 generations had an increased behavioral sensitivity to the F0-conditioned odor, but not to other odors. When an odor (acetophenone) that activates a known odorant receptor (*Olfr151*) was used to condition F0 mice, the behavioral sensitivity of the F1 and F2 generations to acetophenone was complemented by an enhanced neuroanatomical representation of the *Olfr151* pathway. Bisulfite sequencing of sperm DNA from conditioned F0 males and F1 naive offspring revealed CpG hypomethylation in the *Olfr151* gene. In addition, *in vitro* fertilization, F2 inheritance and cross-fostering revealed that these transgenerational effects are inherited via parental gametes. Our findings provide a framework for addressing how environmental information may be inherited transgenerationally at behavioral, neuroanatomical and epigenetic levels.

Functional and structural development

Now let's go back to more “typical” topics in cognitive science.

Definitions

Sensitive period:

It is a limited epoch during development in which the effects of experience on the brain are usually stronger; certain capacities are quickly shaped or altered by experience.

Definitions

Critical period:

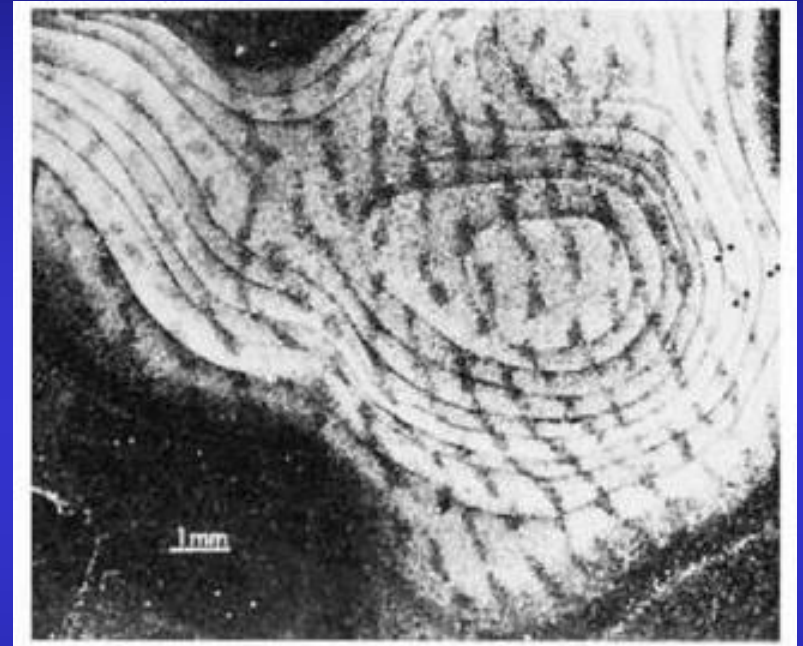
Specific class of sensitive periods that result in irreversible changes in brain functions. After the end of the critical period, effects of atypical experience during the critical period cannot be remediated by restoring typical experience.

Sensitive/Critical periods are reflected in behavior but they actually are a **property of neural circuits!**
(see Hensch et al., 2005)

Cortical changes due to monocular deprivation

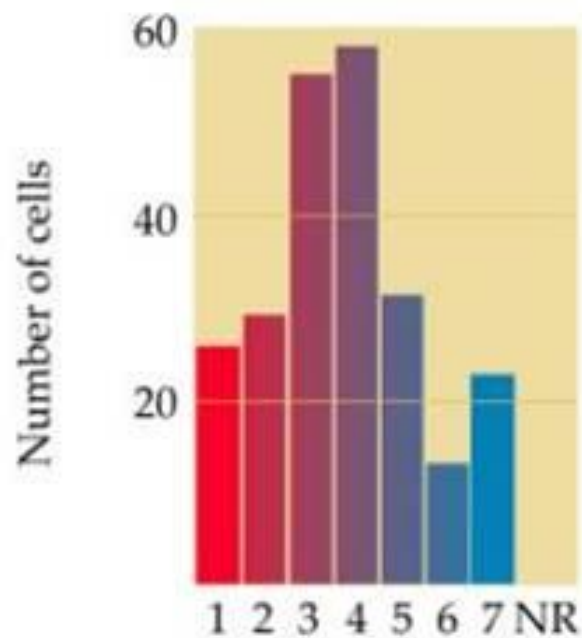


VISUAL CORTEX OF A NORMAL
ADULT MONKEY



VISUAL CORTEX OF A MONKEY
SUBJECTED TO MONOCULAR
DEPRIVATION FROM 2 WEEKS TO
18 MONTHS OF AGE

(A) Normal adult



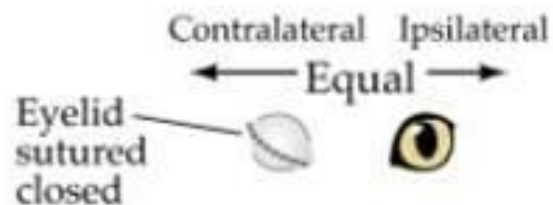
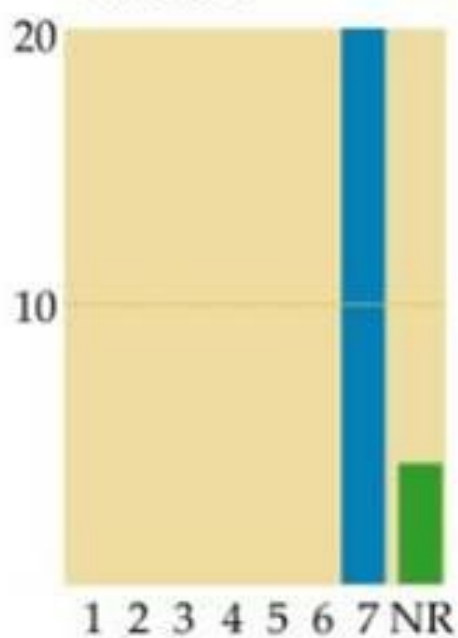
Ocular dominance group



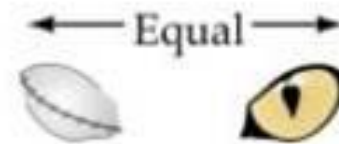
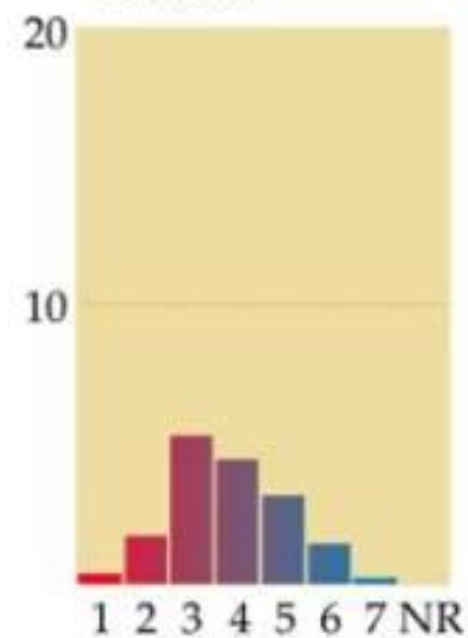
Period of deprivation (months)

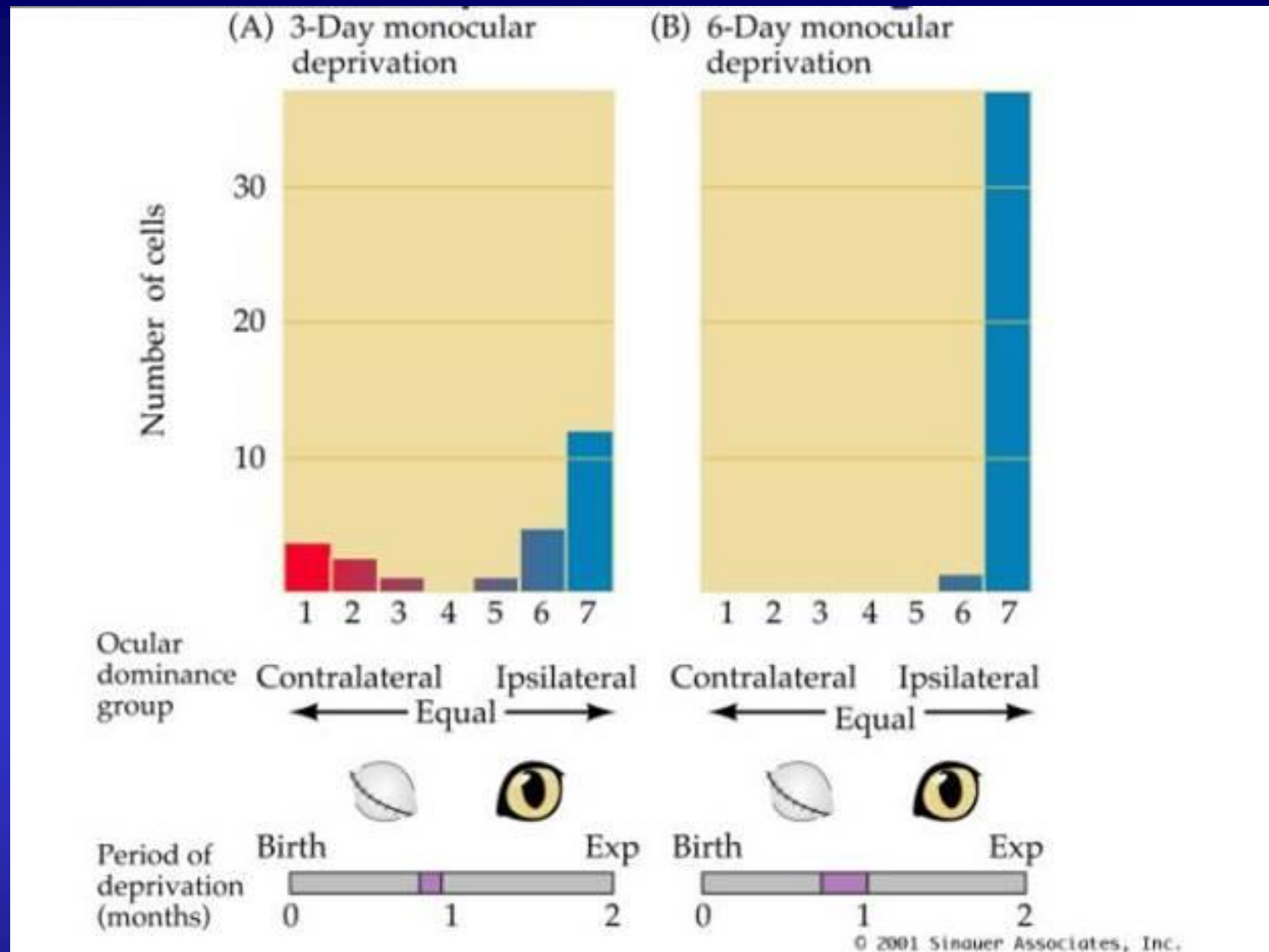


(B) Monocular deprivation in kitten



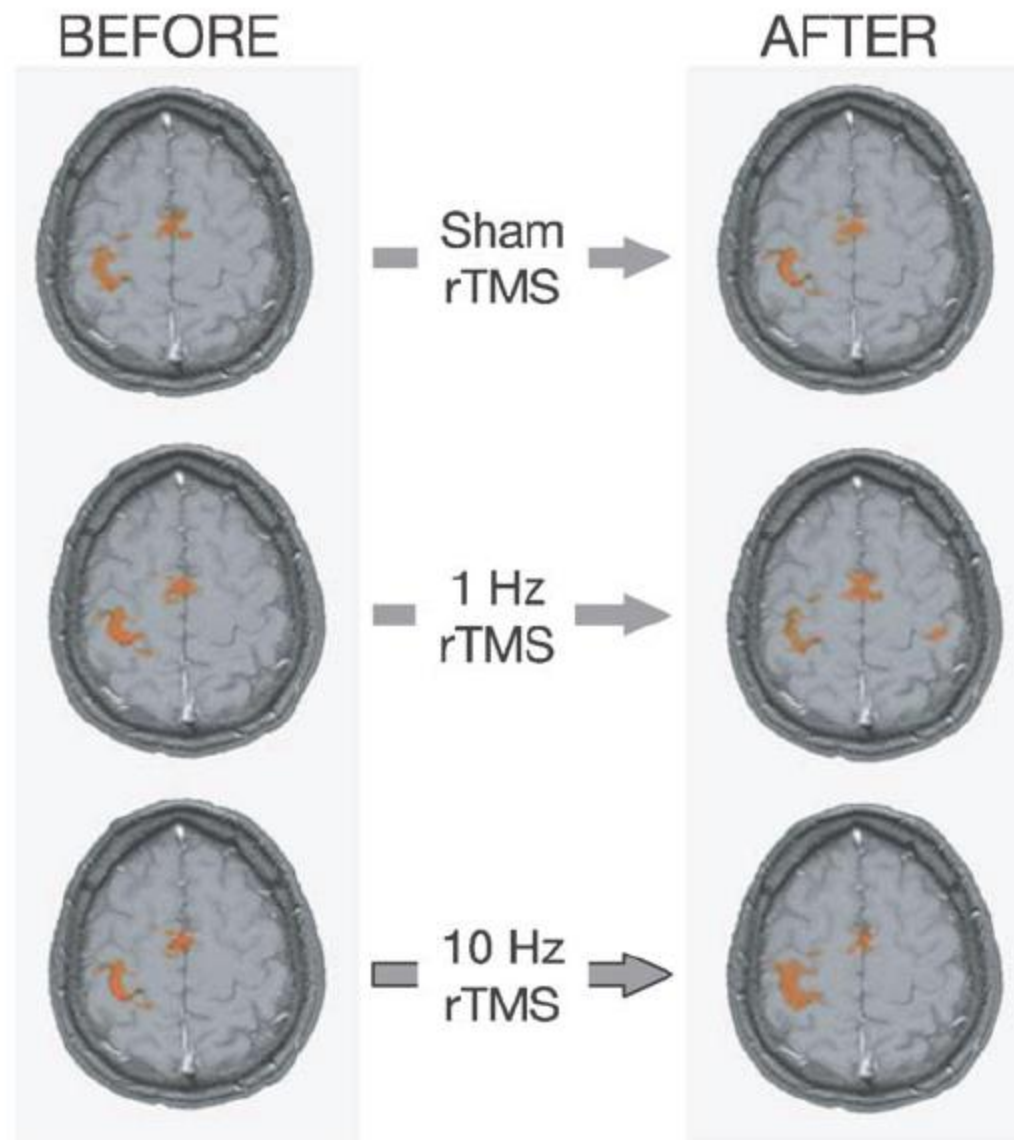
(C) Monocular deprivation in adult



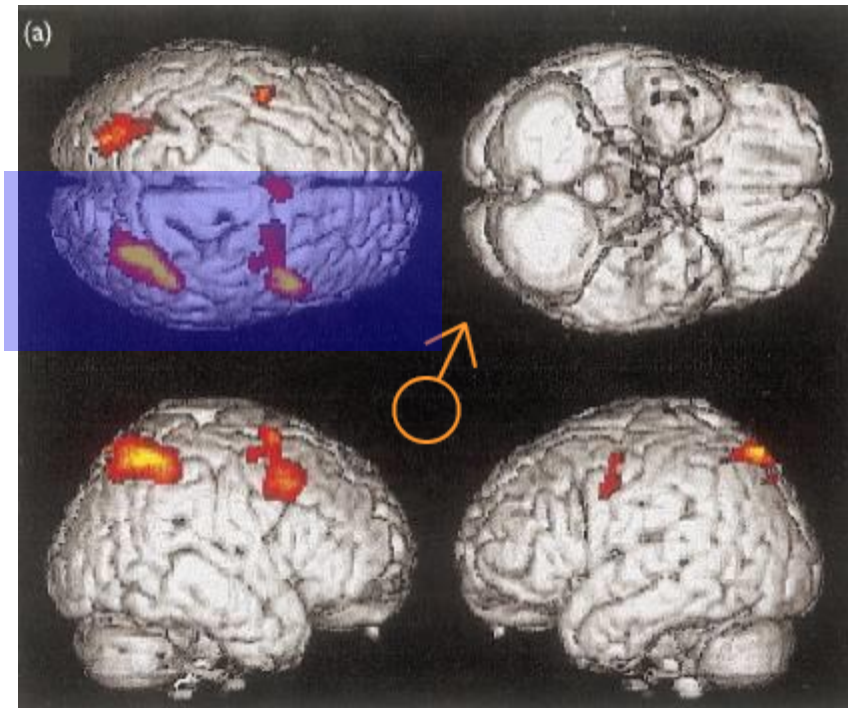


Even just 3-6 days of monocular deprivation could induce major changes in cortical organization

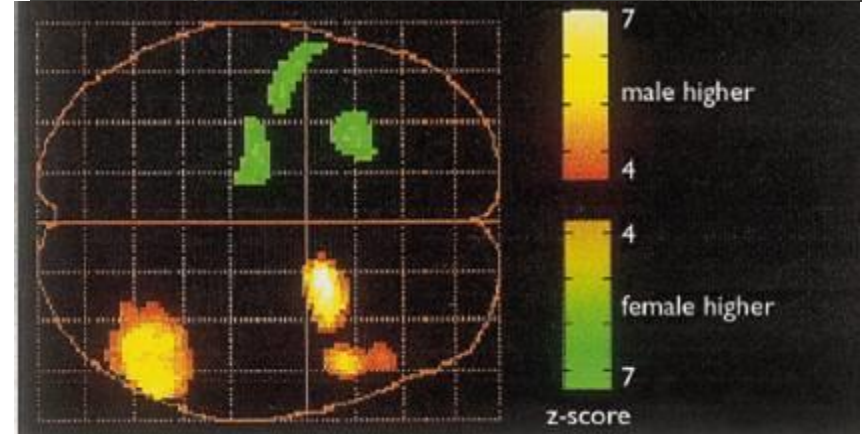
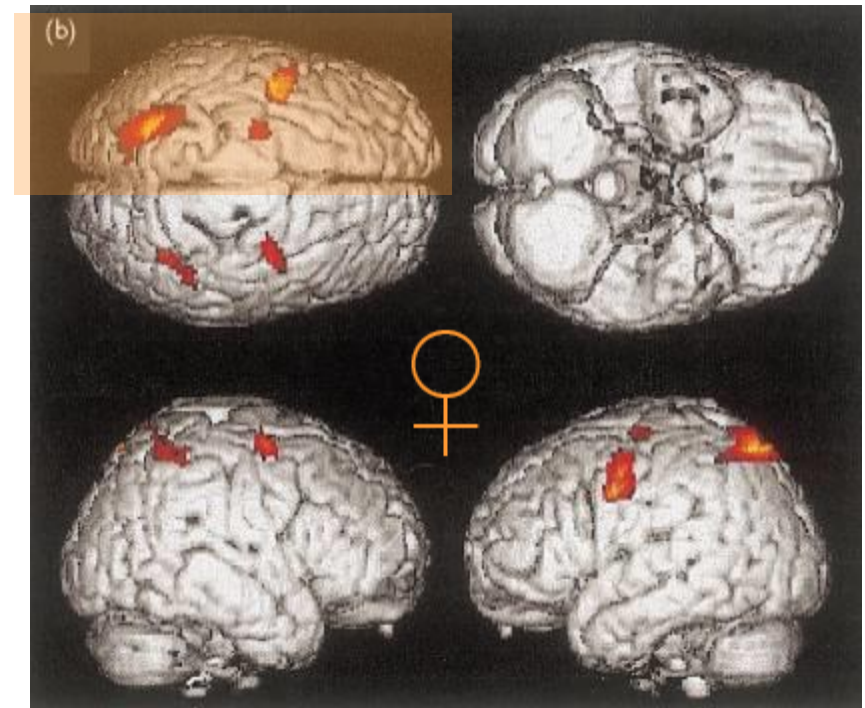
Revision of timing... fast changes!



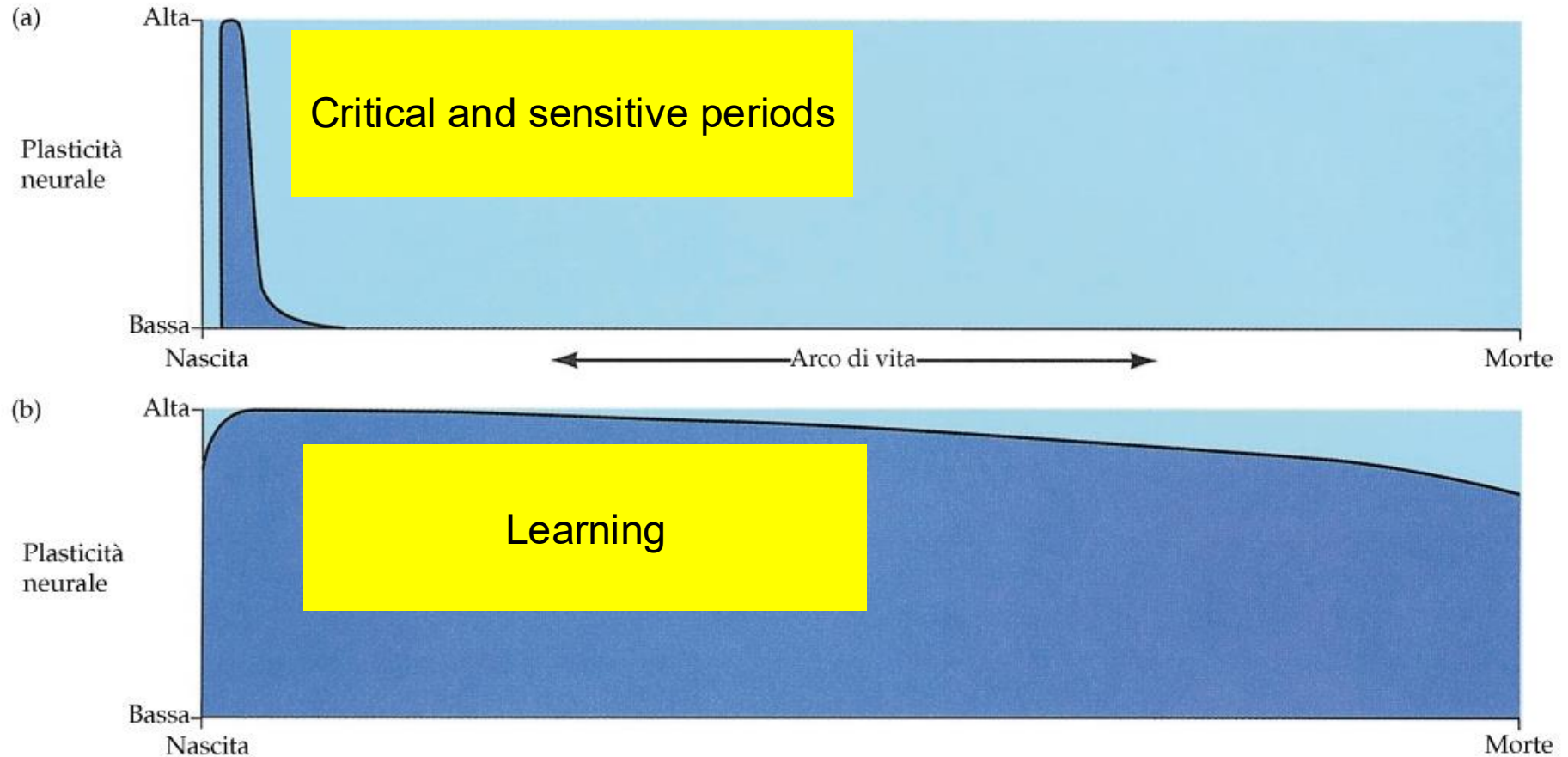
Exogenous modulation of plasticity



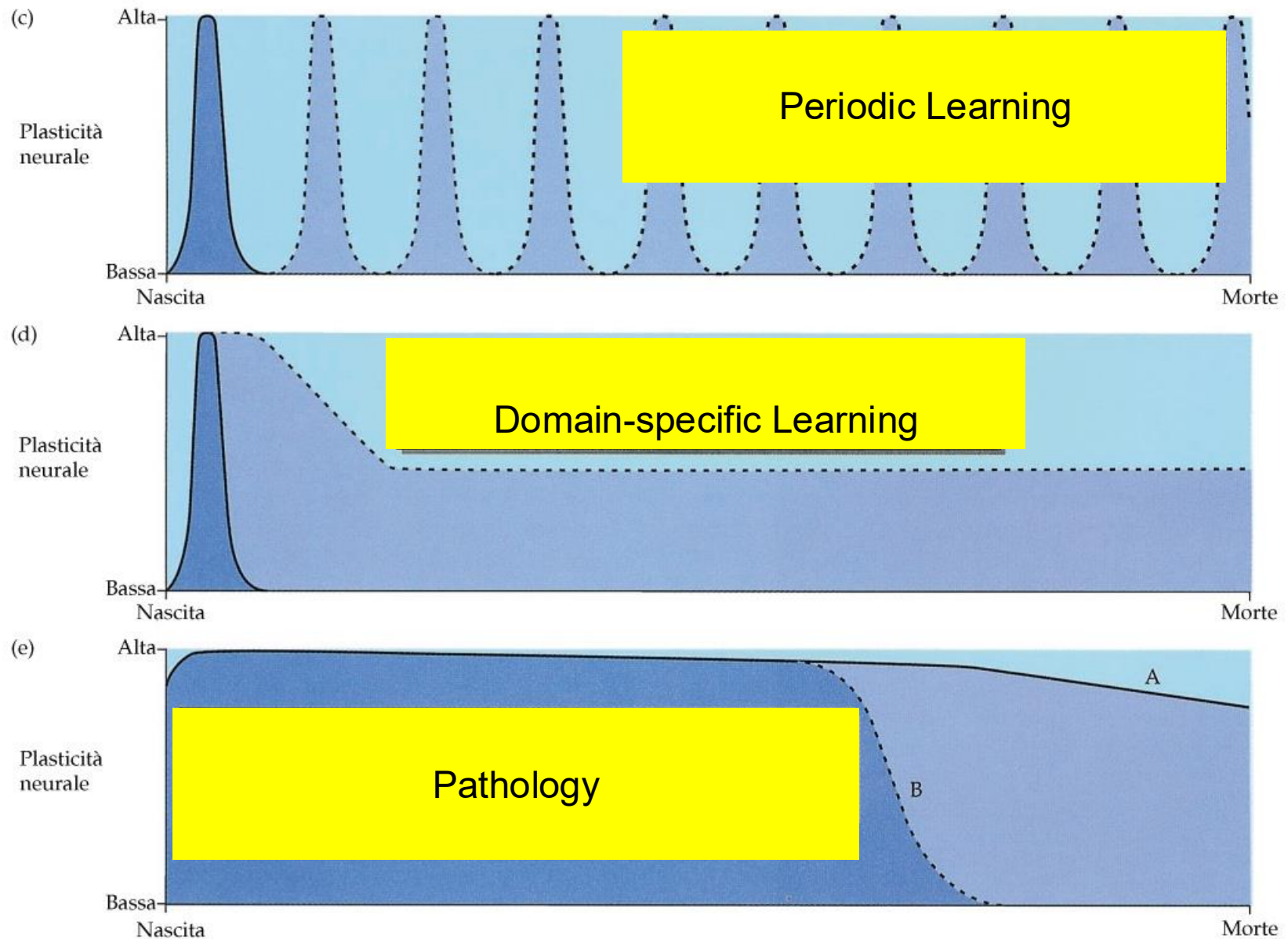
Gender differences in the functional organization of the brain for working memory



Neuronal plasticity changes over the course of life, in different ways depending on the type of task and on the species



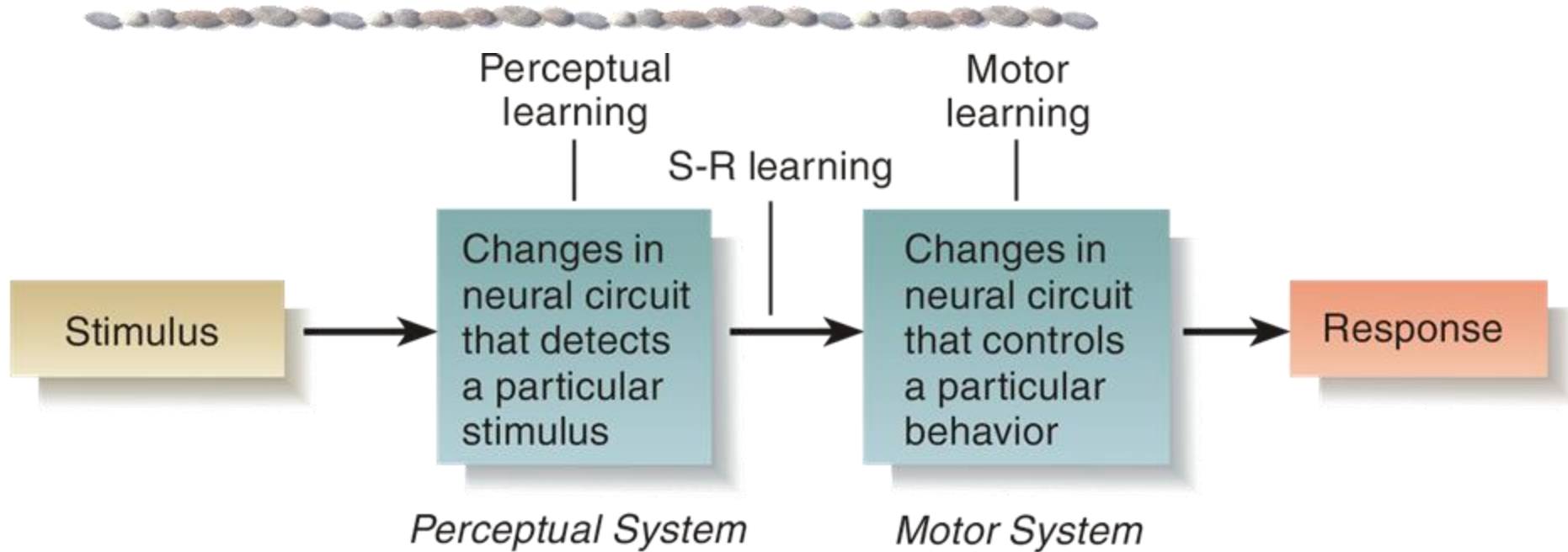
Neuronal plasticity changes over the course of life, in different ways depending on the type of task and on the species



Plasticity: Learning

- 
- “...permanent changes in behavior produced by experience”
 - Learning involves changes in the nervous system that are produced by experience
 - These changes in the nervous system are physical
 - Learning allows us to adapt to the environment
 - Learning requires interactions between the sensory, motor, and memory systems

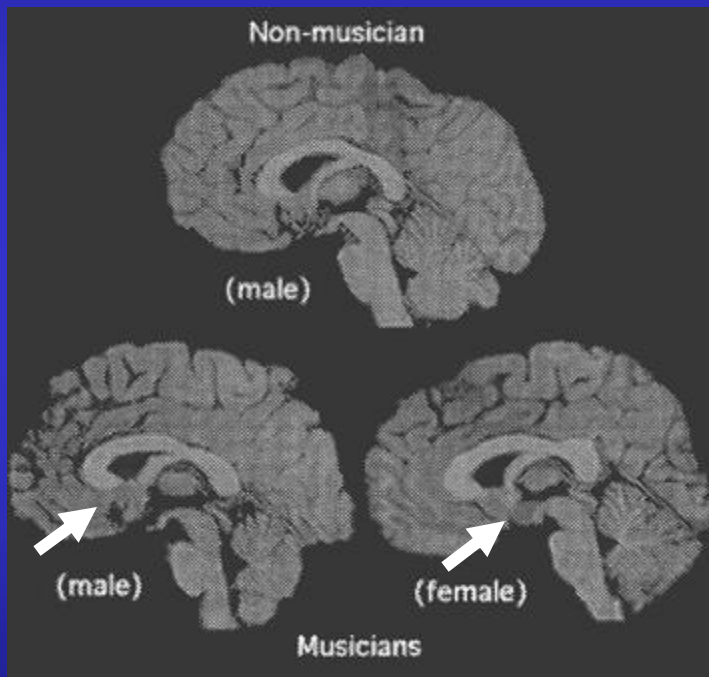
Plasticity: Learning



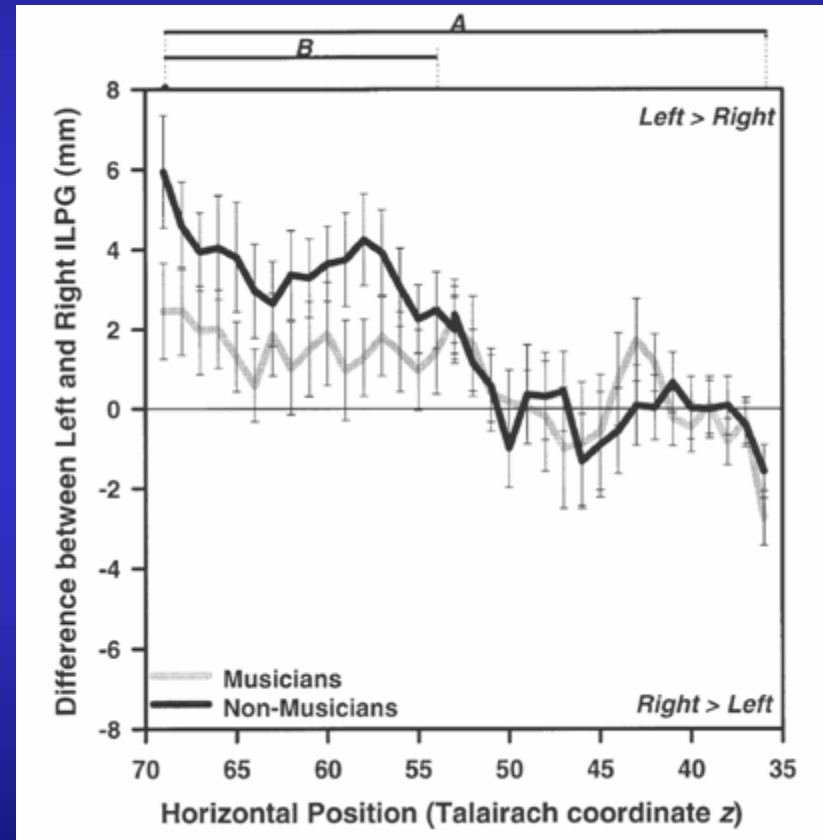
- Perceptual learning: recognizing objects or situations
- Stimulus–response learning: performing an action in response to a specific stimulus (classical or operant conditioning)
- Motor learning: formation of new circuits within the motor system
- Relational learning: identifying relationships or connections among stimuli

The brain of musicians

- Plastic changes associated with the learning of specific motor patterns



Increase size of anterior commissure



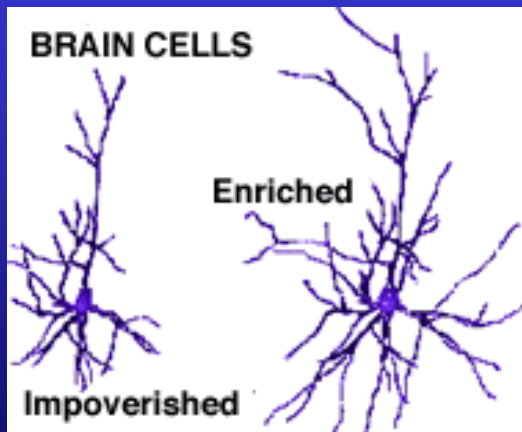
Anatomical changes of motor cortex

Brain development and refinement

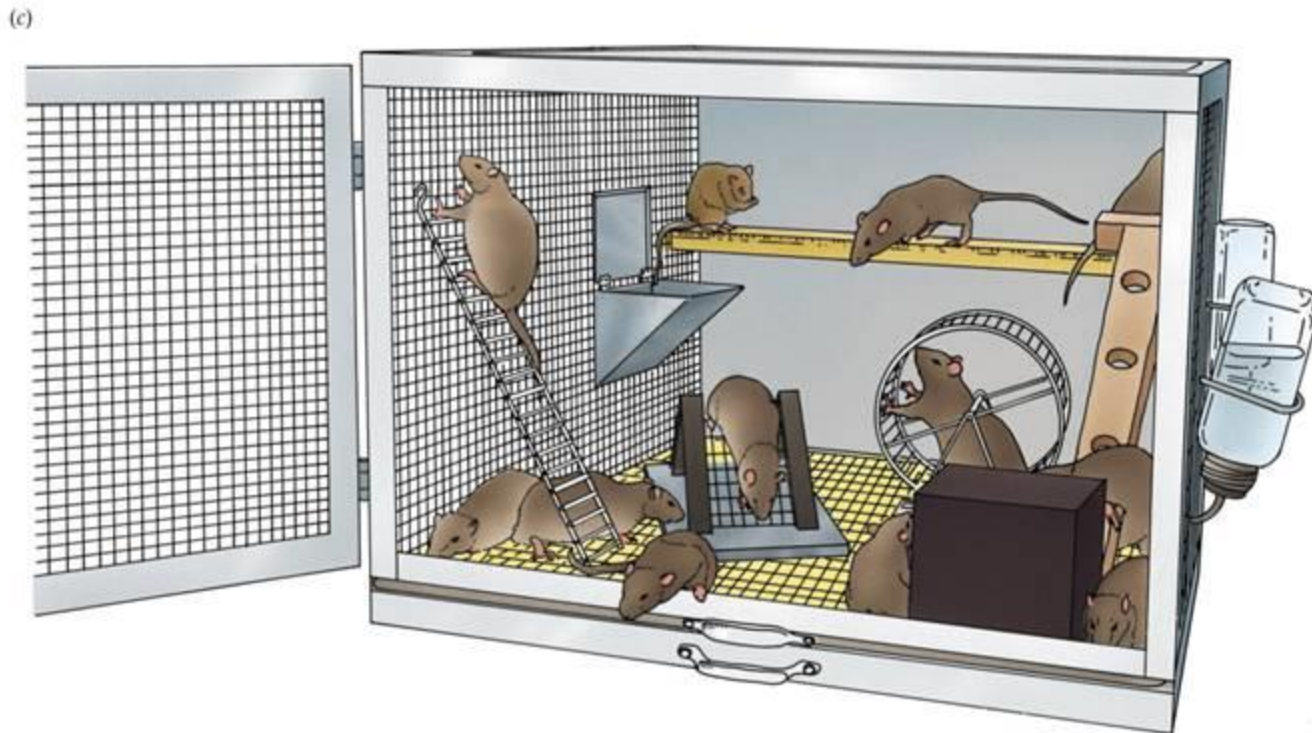
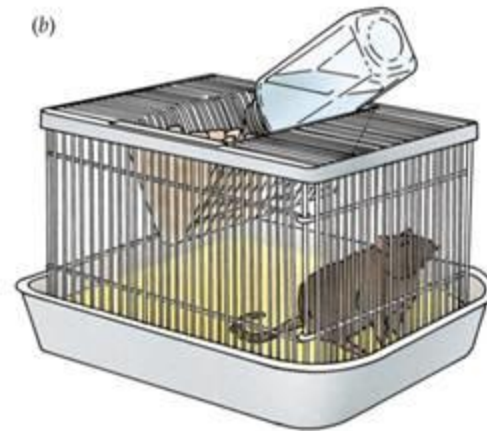
- The anatomical differentiation of the brain is closely tied to experience.
 - Nature provides the basic processes, while the environment provides the experiences that select the most appropriate neural networks based on use and function.
- Some connections are strengthened and become more defined through use, while many others regress or disappear.
- Brain maturation is an organized and hierarchical process

Brain development and refinement

- Effects of experience on dendritic arborization
 - Changes in cortical areas
 - The brain of rats raised in an enriched environment shows:
 - an increase in enzymatic activity, particularly AChE
 - greater weight and thickness of the cerebral cortex, especially in the occipital region, where a 20% increase in dendritic spines, branching, and synaptic contacts is observed

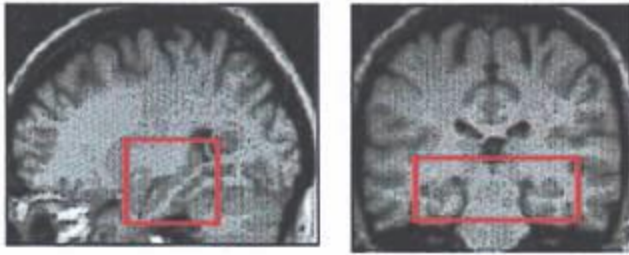


Brain development and refinement

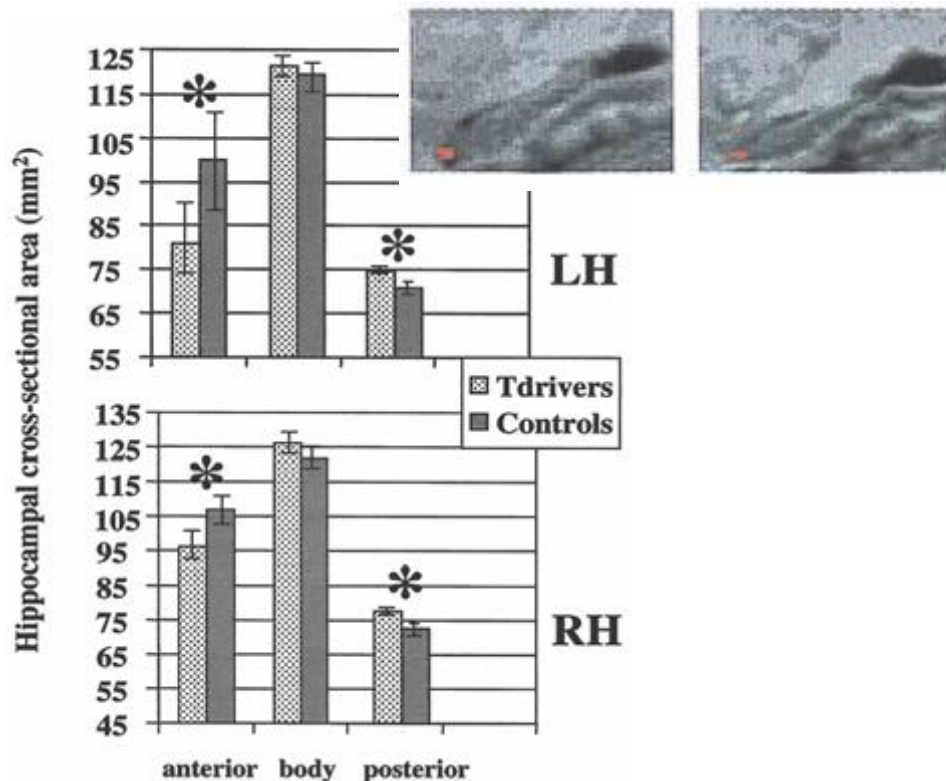


Navigation-related structural change in the hippocampi of taxi drivers

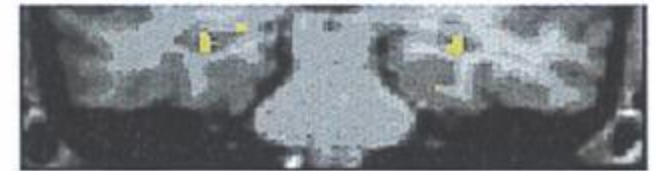
Eleanor A. Maguire[†], David G. Gadian[†], Ingrid S. Johnsrude[†], Catriona D. Good[†], John Ashburner[†], Richard S. J. Frackowiak[†], and Christopher D. Frith[†]



Anterior hippocampus: > control group



Posterior hippocampus and body: > taxi drivers



y = -33



y = -27

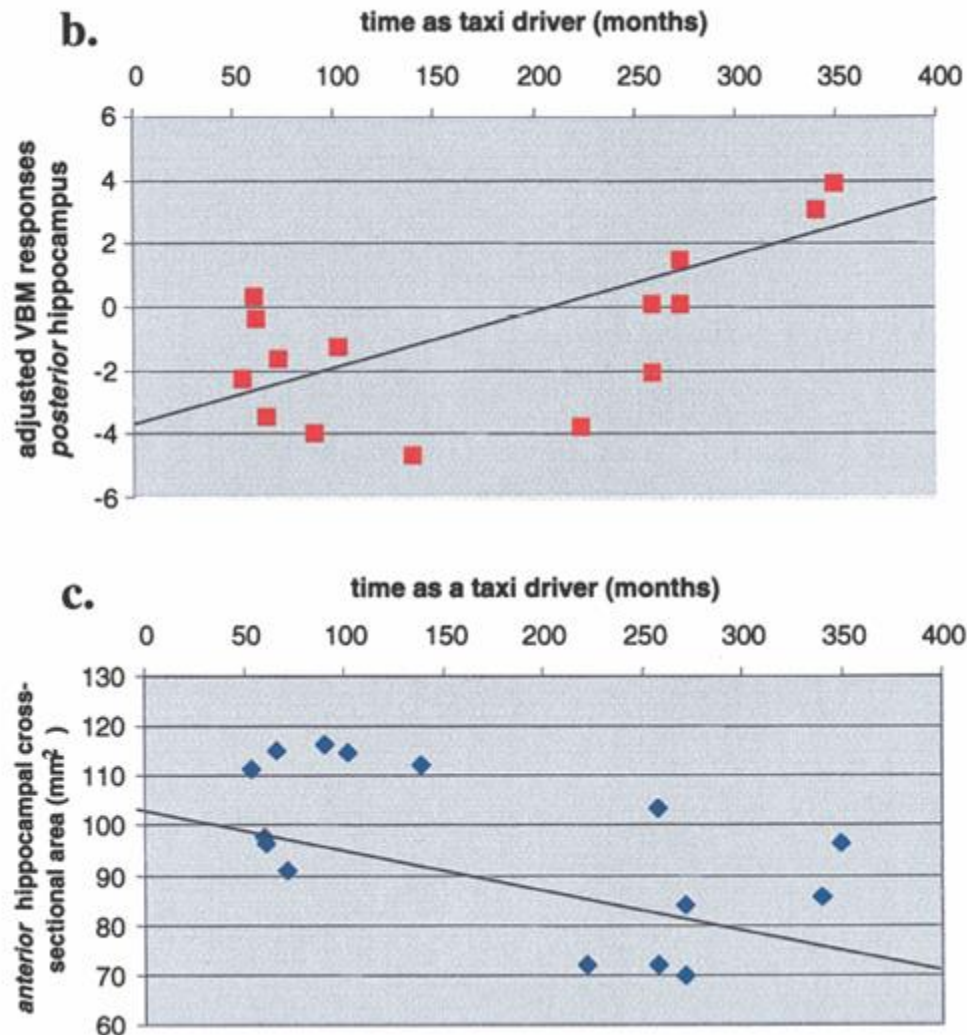


y = -20

Navigation-related structural change in the hippocampi of taxi drivers

Eleanor A. Maguire^{*†}, David G. Gadian[†], Ingrid S. Johnsrude[†], Catriona D. Good[†], John Ashburner[†], Richard S. J. Frackowiak[†], and Christopher D. Frith[†]

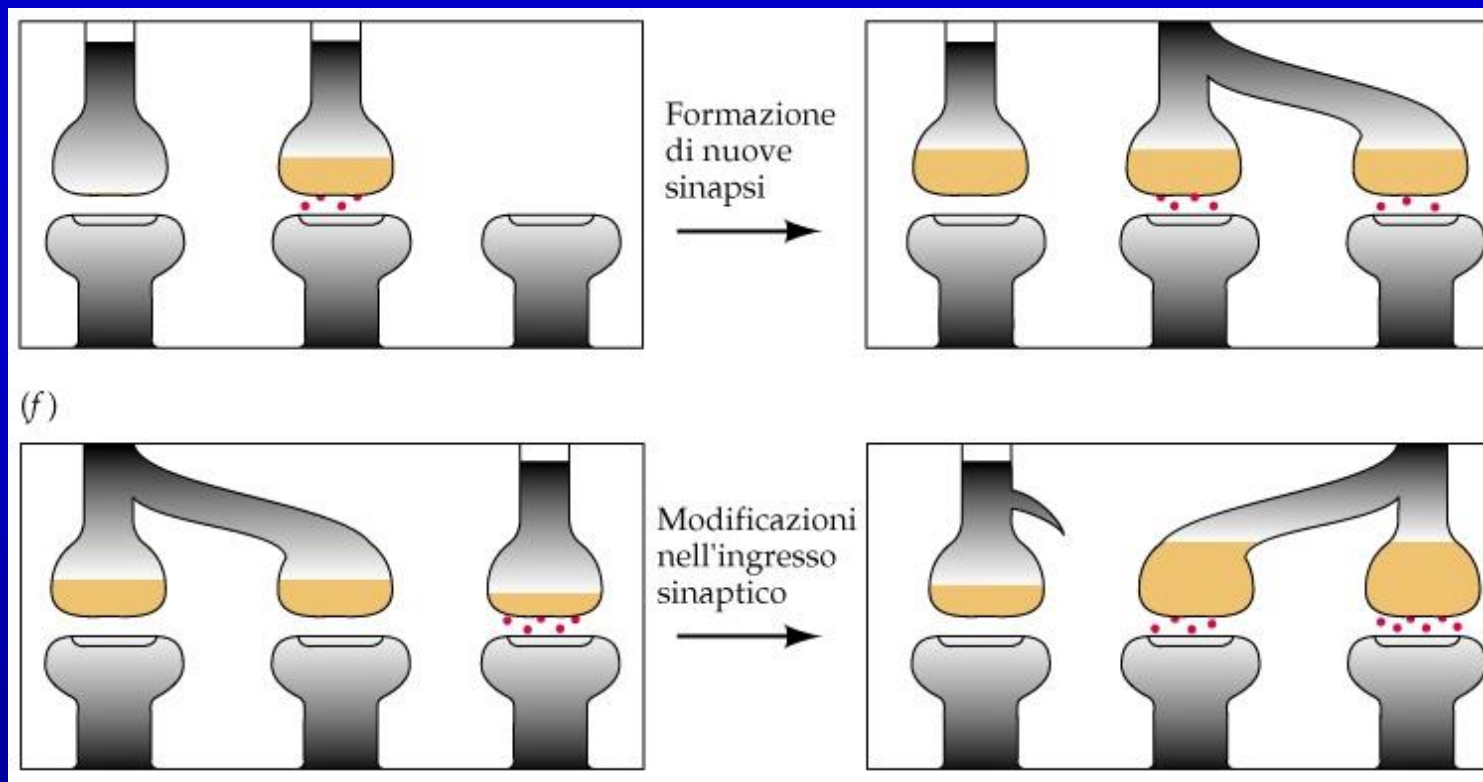
It therefore appears that adaptive changes occur in the posterior part of the hippocampus (involved in storing spatial representations of the surrounding environment), shaped by the specific demands of the environment.



Post-damage plasticity

- **sprouting**

- Axons can take the place of vacant synapses after neighbouring neurons die
- A neuron can become responsive to other axons if the axons that originally innervated it become inactive or die



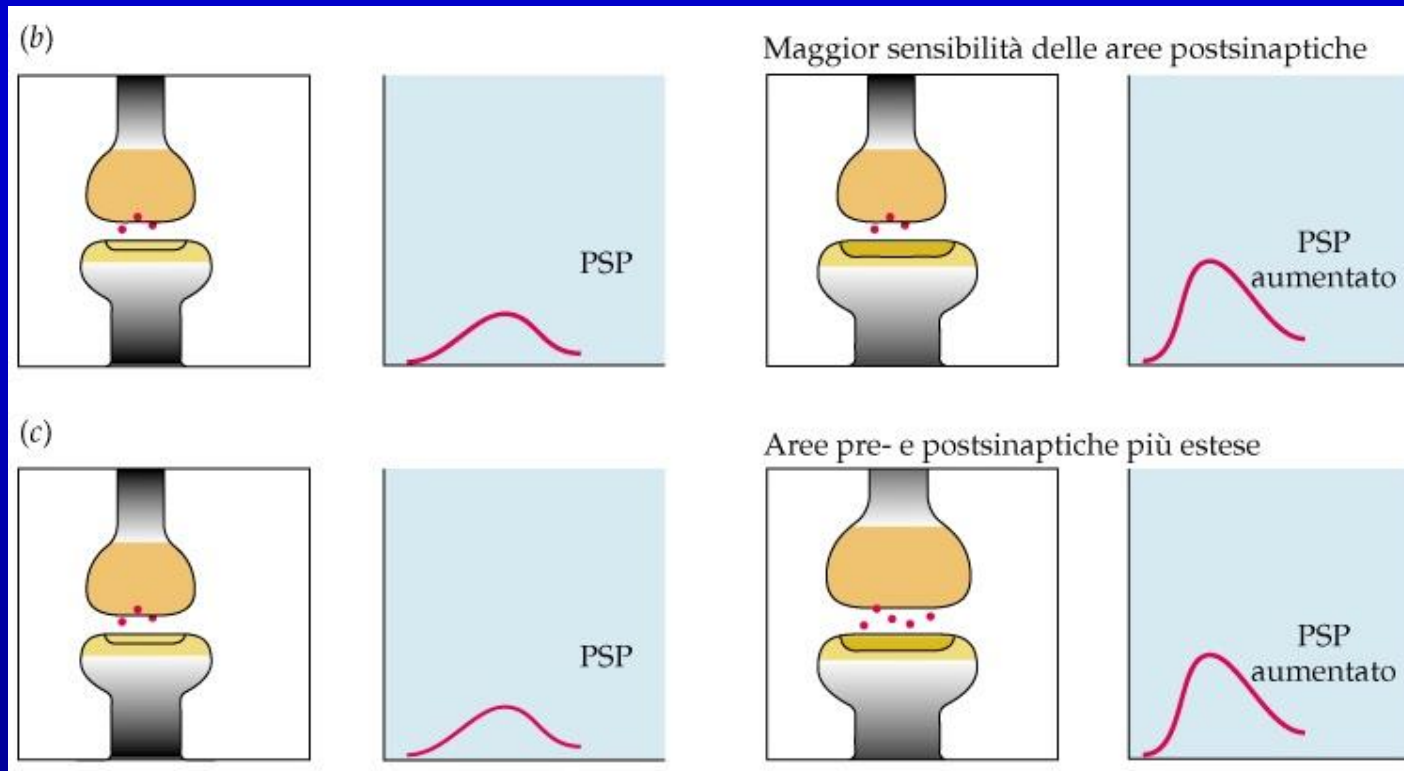
Plasticità dello sviluppo

- **Denervation supersensitivity**

- Increased response to a neurotransmitter following the destruction of the afferent axon

- **Inactivity-induced supersensitivity**

- Increased response to a neurotransmitter due to inactivity of the afferent axon

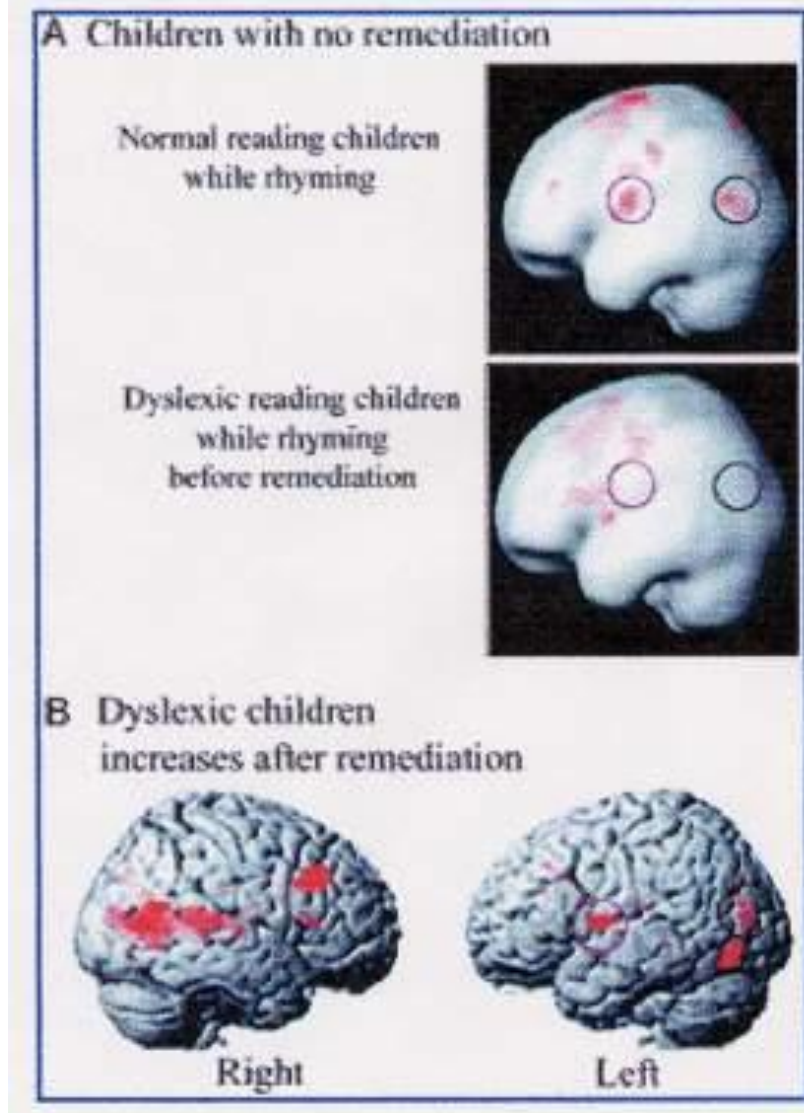


Plasticity



- **Compensatory masking** following injury or learning, an implicit or explicit change in task-execution strategy occurs, allowing a secondary cognitive process—normally playing a minor role—to assume a primary role)
- **Expansion of the functional map** following injury, a functional region adjacent to the damaged area expands its functional/neural territory into neighboring cortex. Learning can also induce an enlargement of the functional extension of a region into adjacent cortical areas
- **Adaptation of homologous regions** following injury, functionally homologous regions in the contralateral hemisphere can compensate for the lost functions
- **Cross-modal reorganization** following sensory deprivation, primary and secondary sensory areas dedicated to that perceptual modality reorganize to respond to other sensory modalities

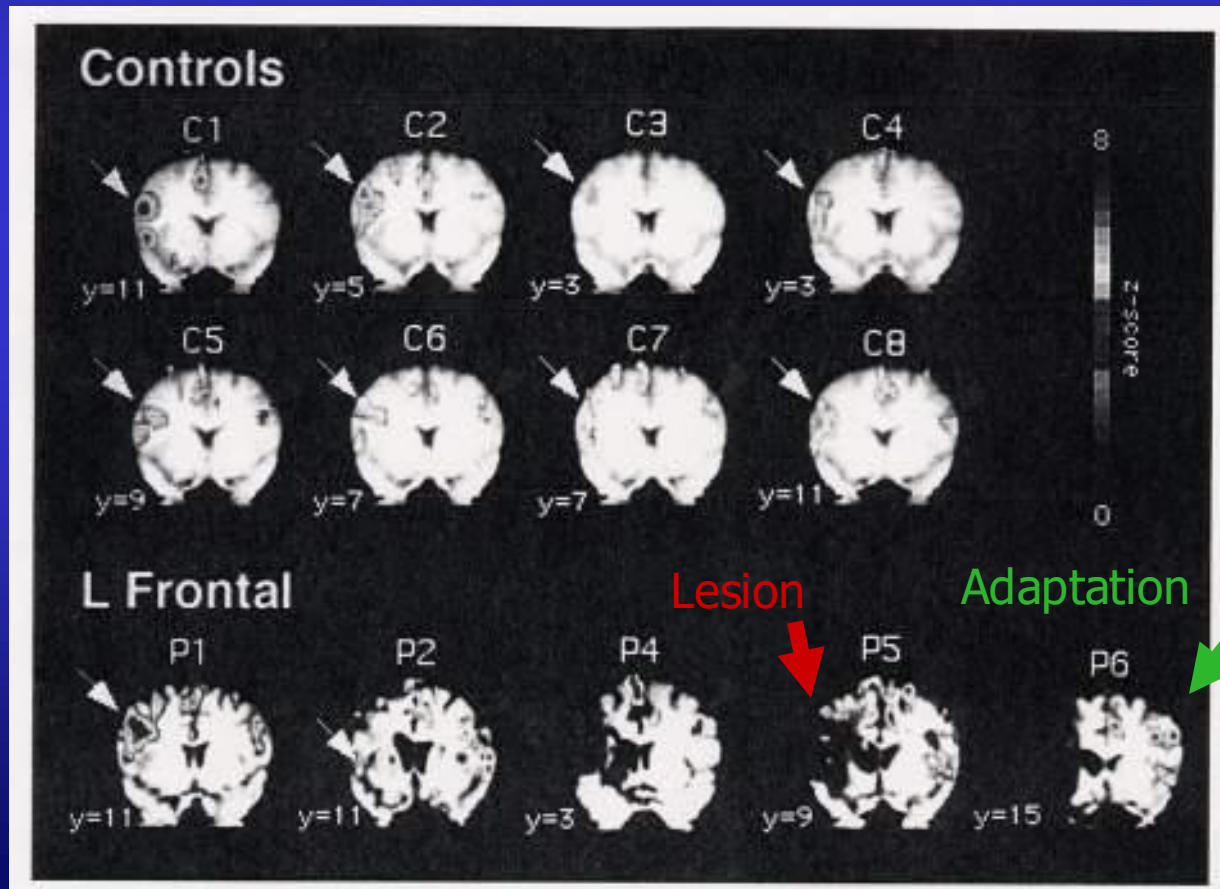
Compensatory masking and adaptation of homologous regions



Phonological training in children with dyslexia induces greater activation in critical language regions in both the left and right hemispheres.

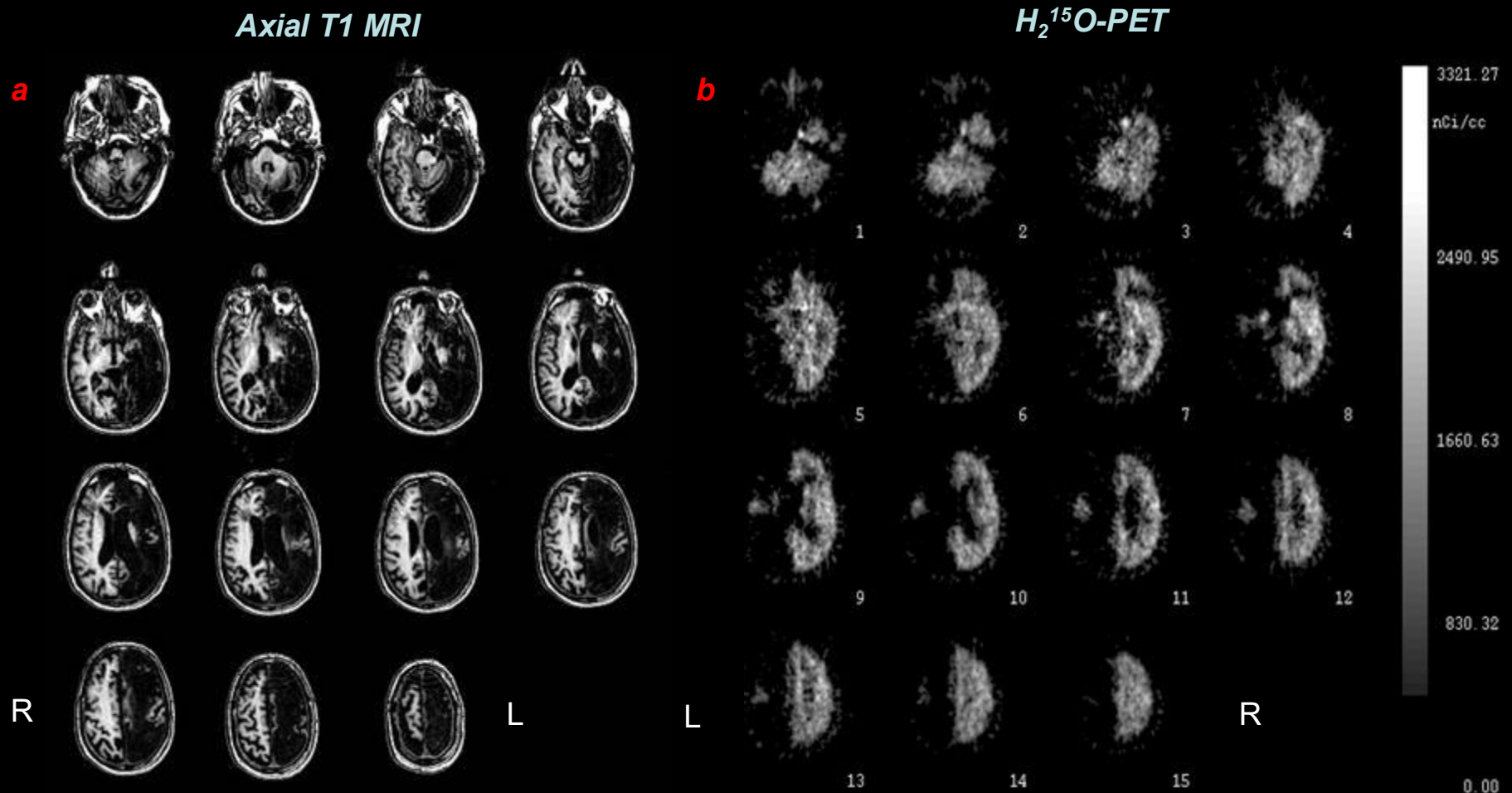
Adaptation of homologous regions

- The regions homologous to the left-hemisphere language areas, located in the right hemisphere, are in some way already prepared to complement language processing: this underlies recovery from aphasia.



PET and fMRI in the Study of Functional Plasticity in the Human Brain

A 67 y/o right-handed male survived a massive stroke of the left hemisphere at age 47 yr. In the left hemisphere, structure (a) and metabolic activity (b) were preserved in limited areas only. We studied potential compensatory effect of the R hemisphere (brain plasticity) in carrying out activities normally sustained by the L hemisphere.

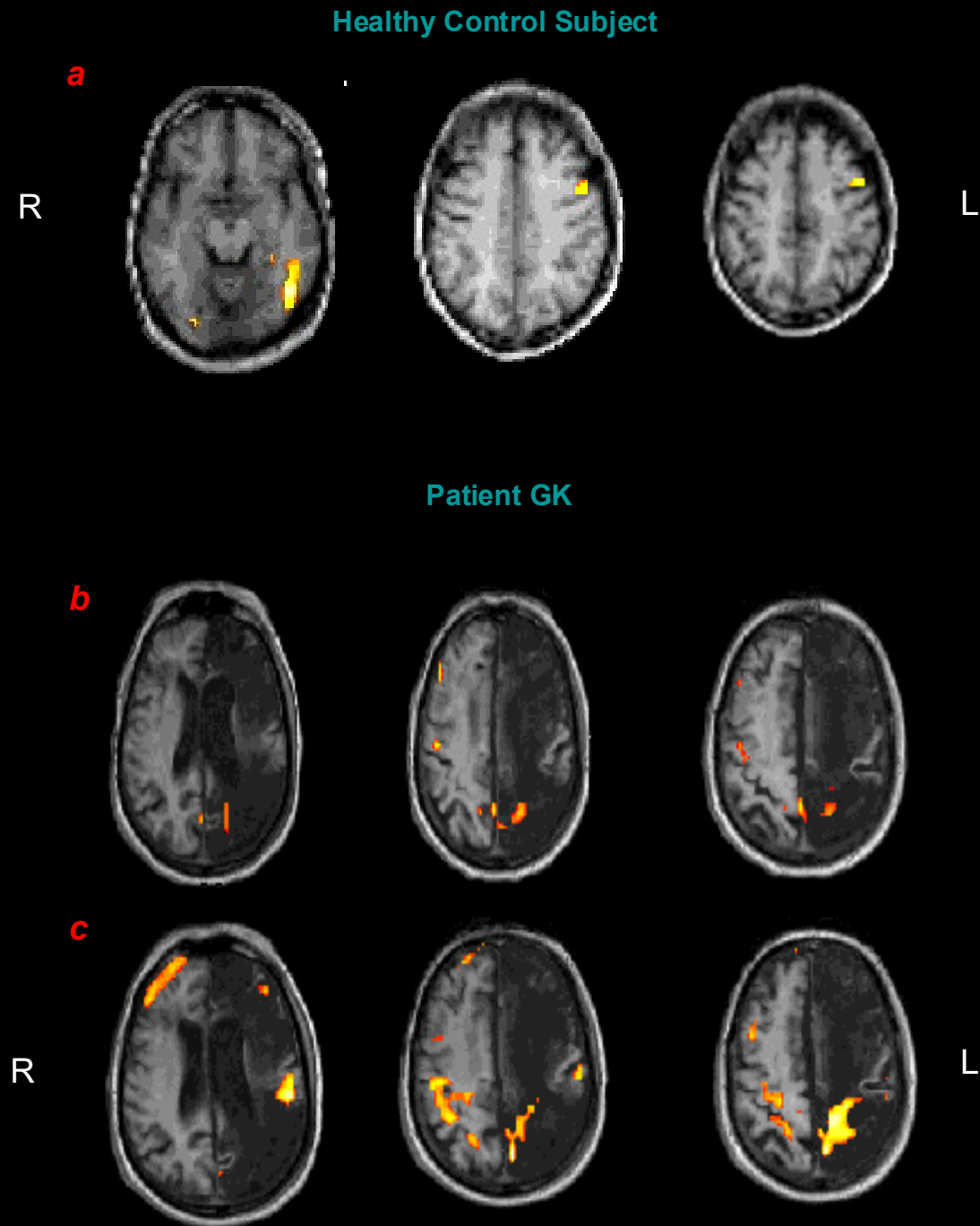


Normally **mathematical functions** rely on parietal and frontal areas of the left hemisphere (a).

In patient GK performance on mathematical tasks was highly compromised and associated with limited activation of residual parietal areas in the left hemisphere (b).

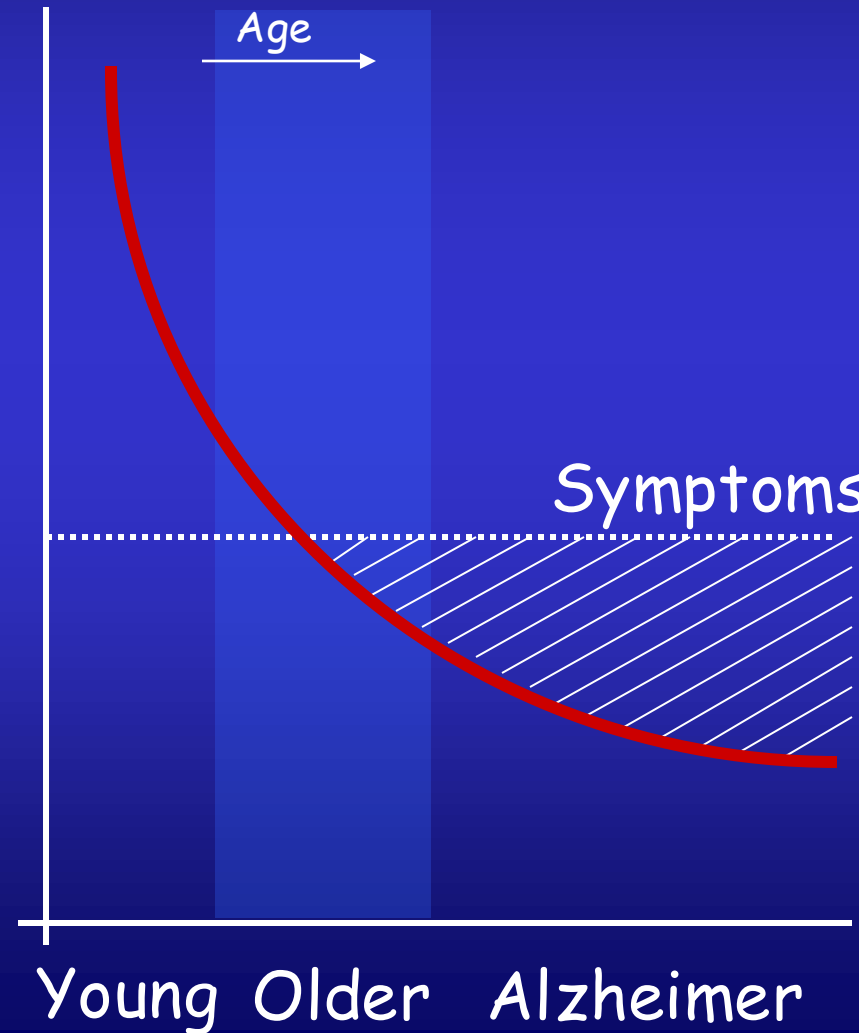
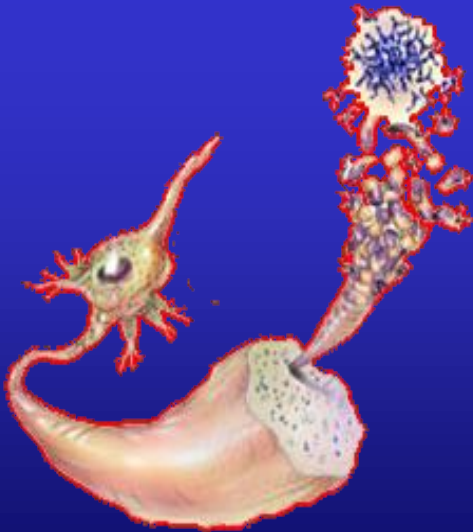
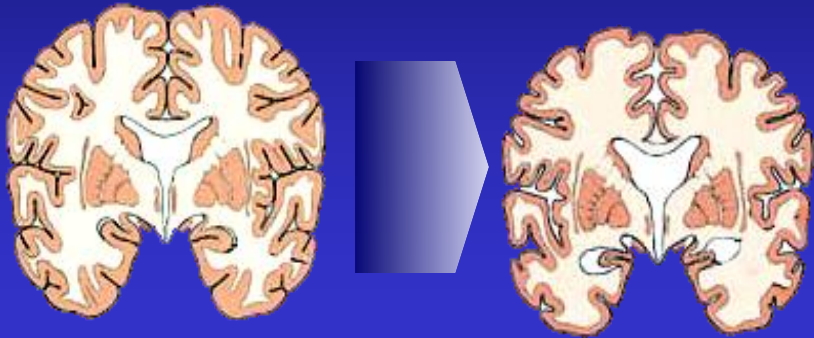
After a period of training the patient reacquired some ability to perform simple mathematical operations. Improved mathematical performance was associated with activation of **homologous parietal and frontal areas in the right hemisphere** (c).

These findings indicate a compensatory process through functional plasticity in the adult brain.



Aging

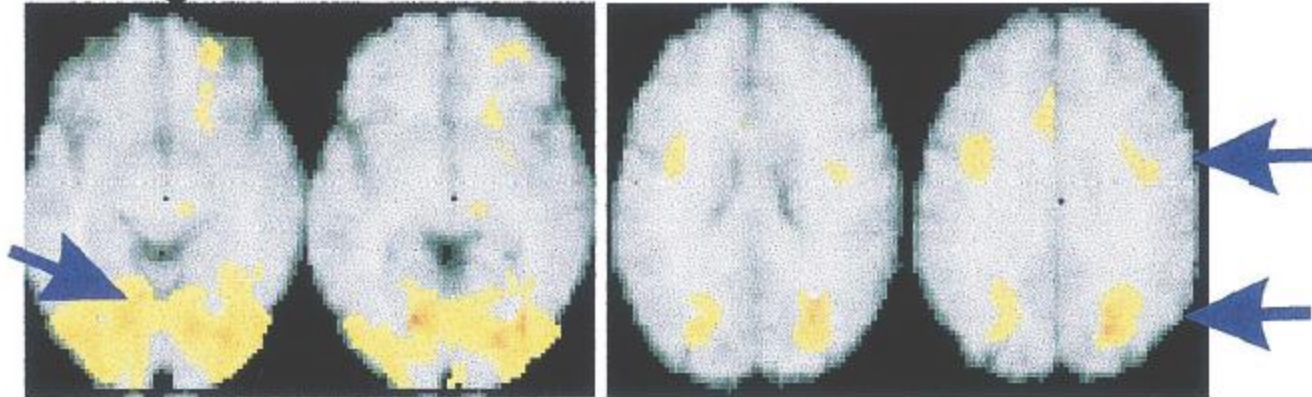
Atrophy and
neuronal loss



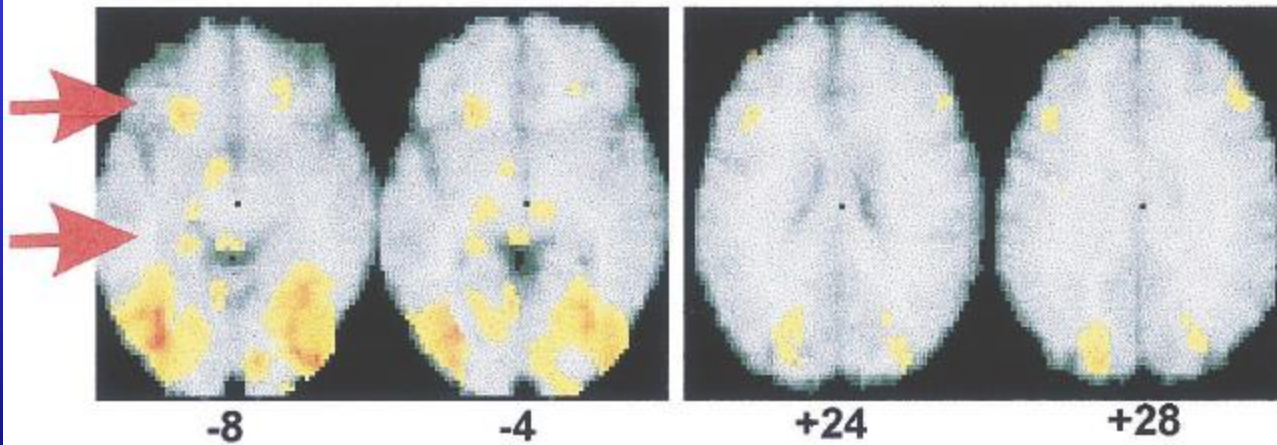
Plasticity during aging

Face Perception

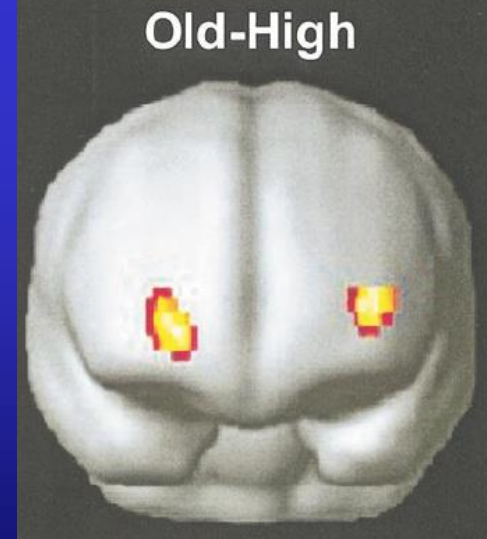
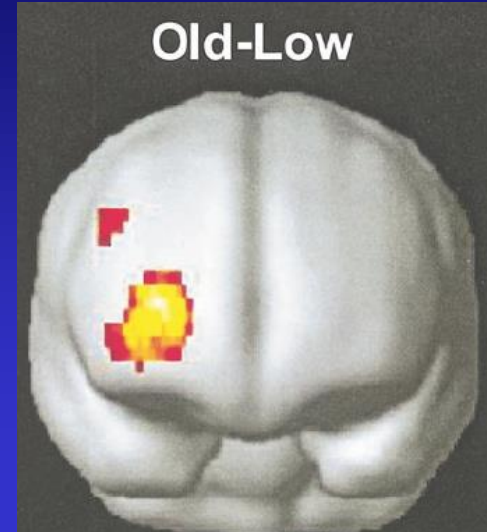
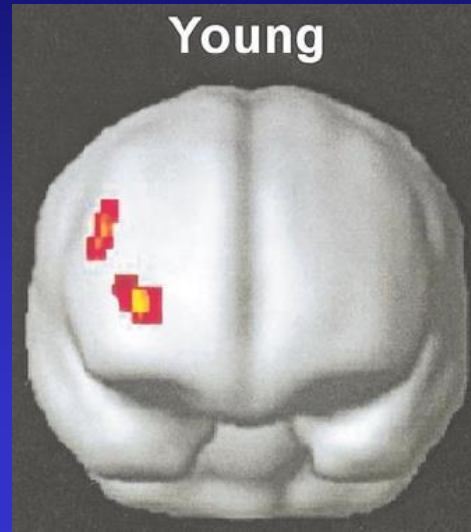
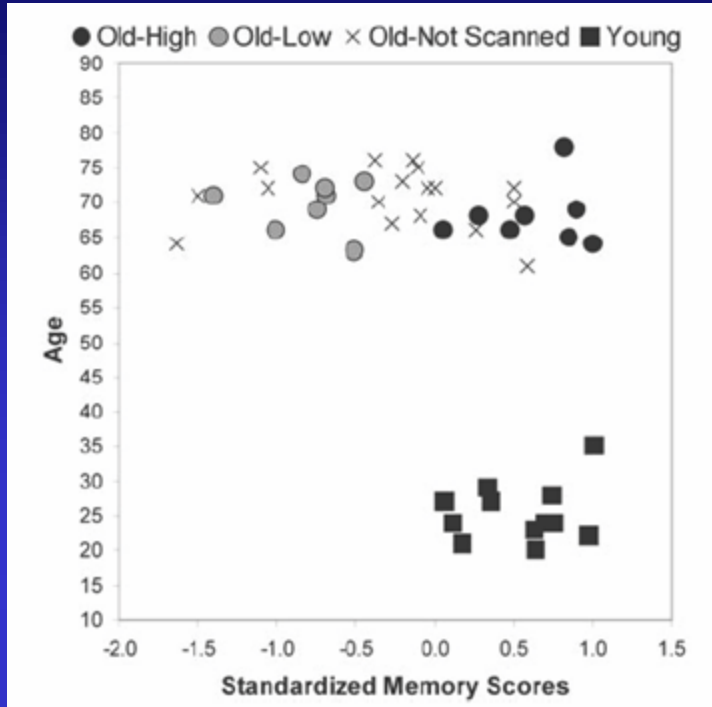
Young



Old



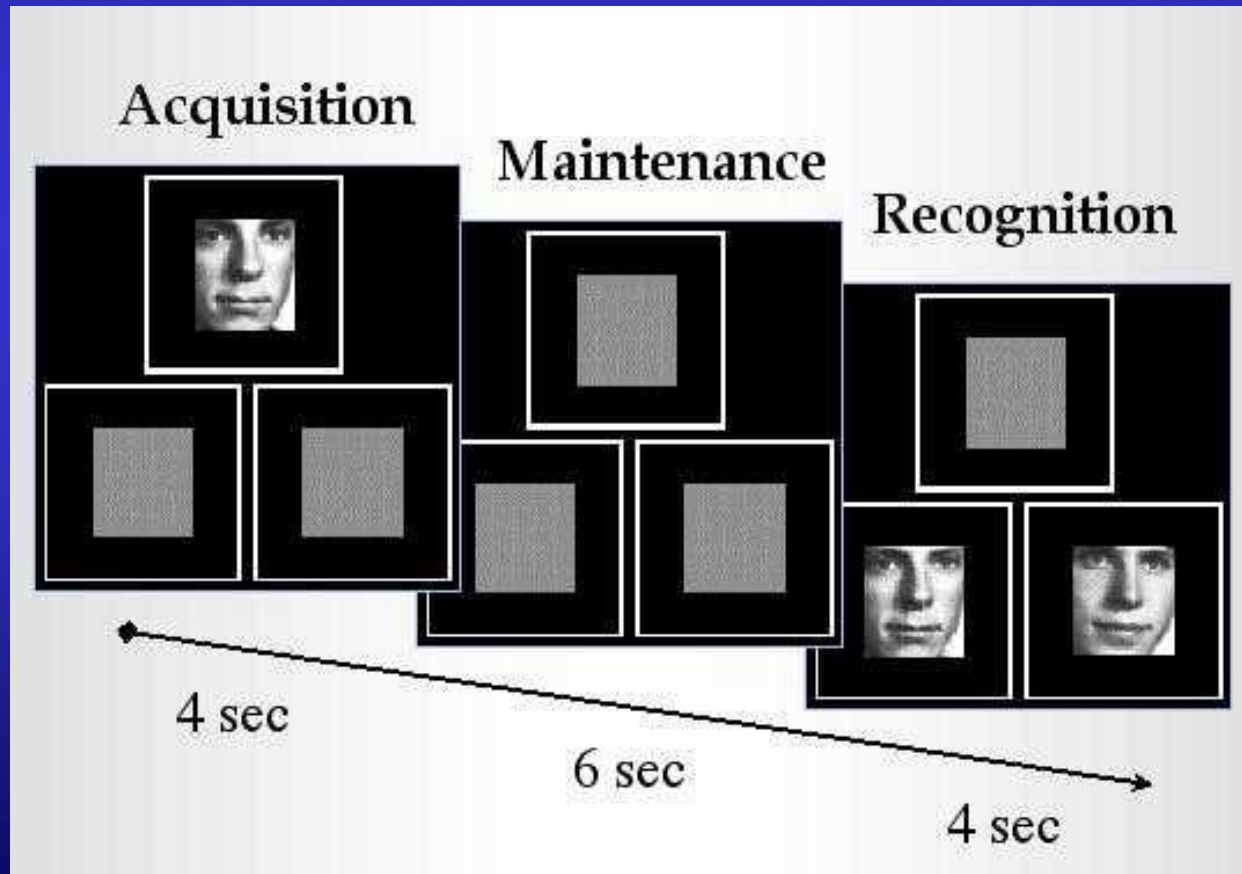
Plasticity during aging



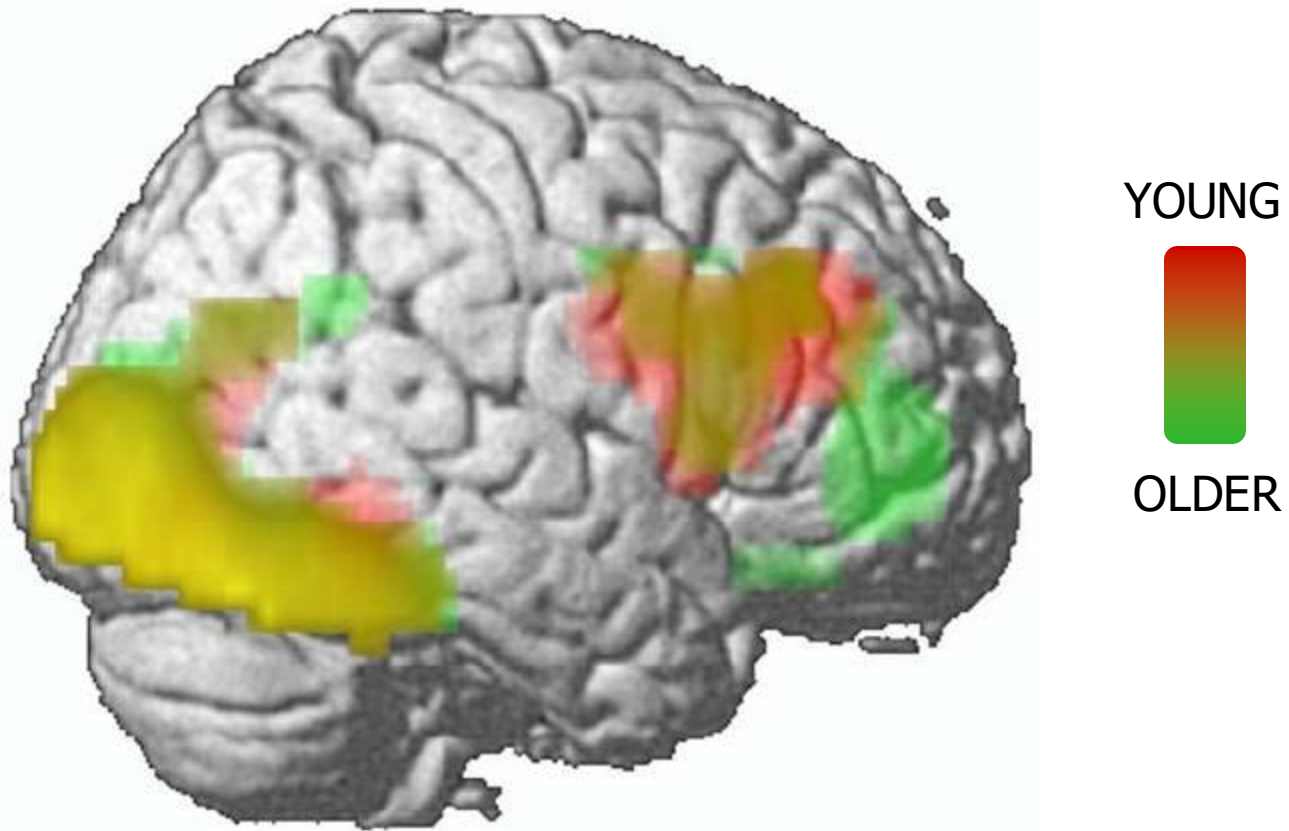
- Distinct brain areas can be “recruited” to compensate for or substitute impaired cognitive abilities.
 - Cortical regions involved in the task
 - Cortical regions tend to compensate when the task becomes more challenging

Plasticity during aging: working memory

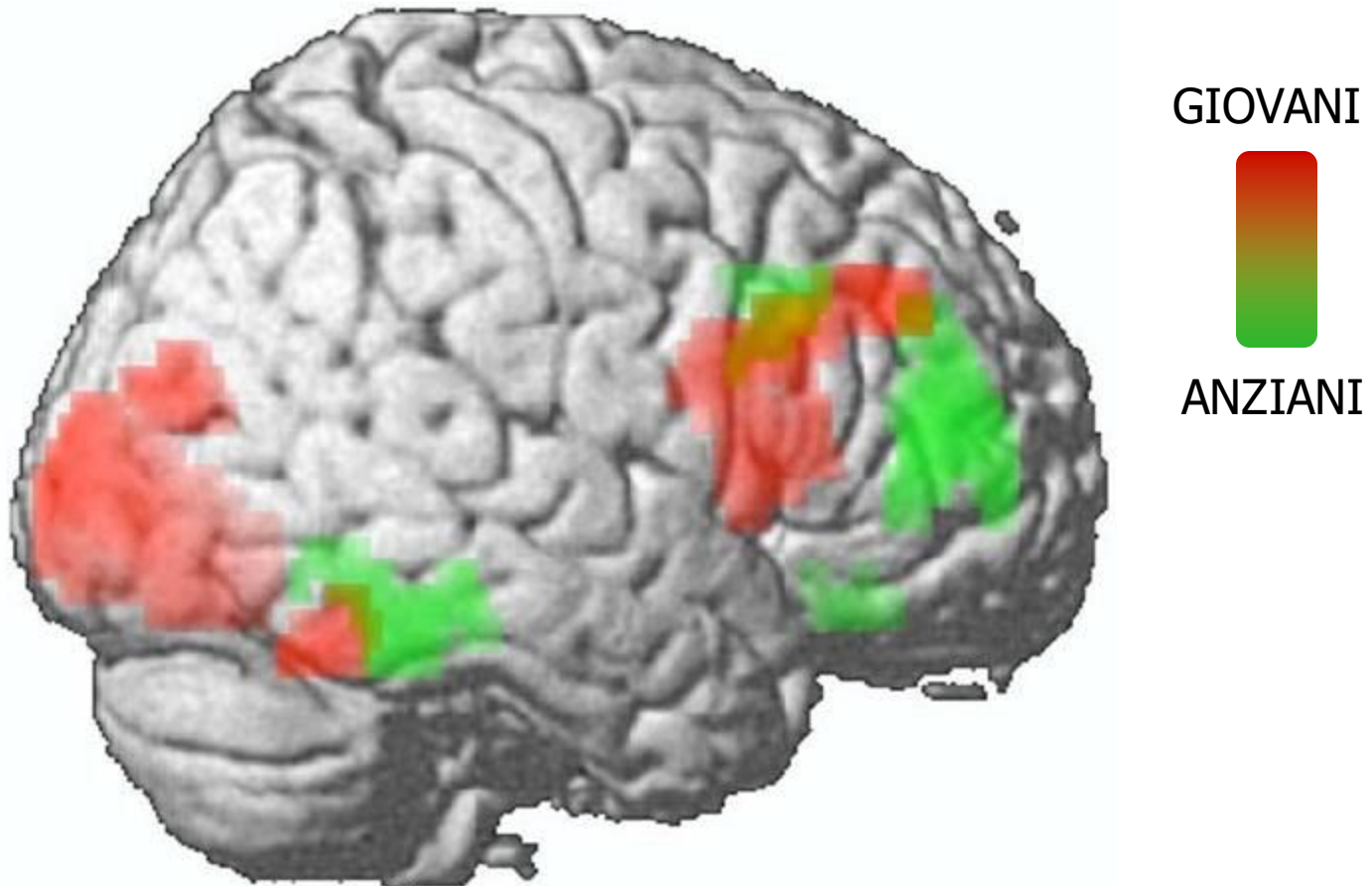
A process that keeps information temporarily active for further processing or recall



Different patterns of cortical recruitment during a working memory task in young and older adults



The administration of a drug that enhances the cholinergic system (physostigmine) specifically affects those cortical regions that were selectively recruited by the two groups before the drug was administered, indicating effective compensation.

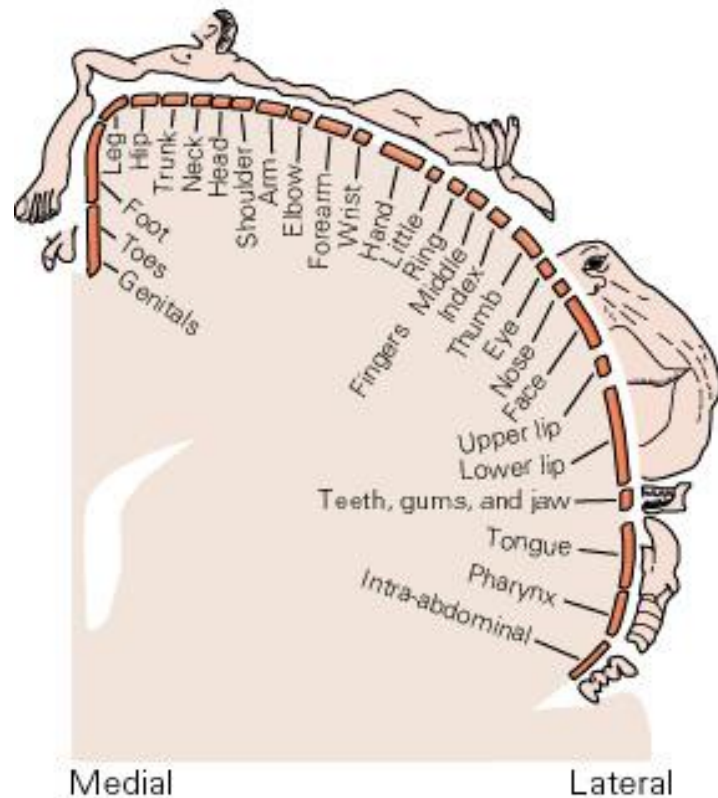


Expansion of the functional map

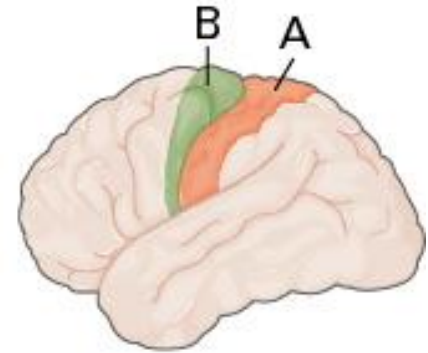
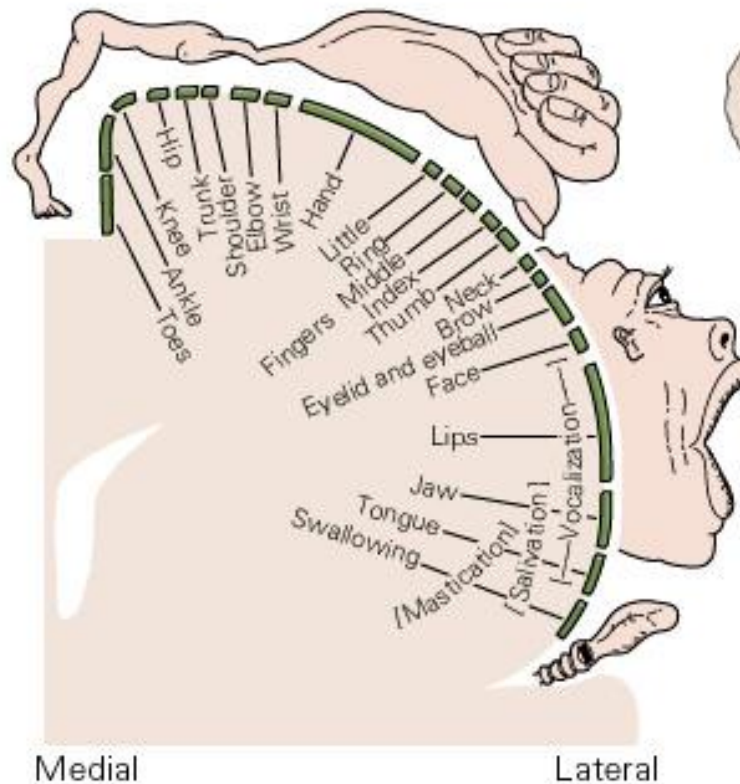
- Following a lesion, a functional region located near the damaged area can expand its functional/neural territory into the adjacent cortex.
- Learning can also induce an expansion of the functional extent of a region into neighboring cortical areas.

Expansion of the functional map

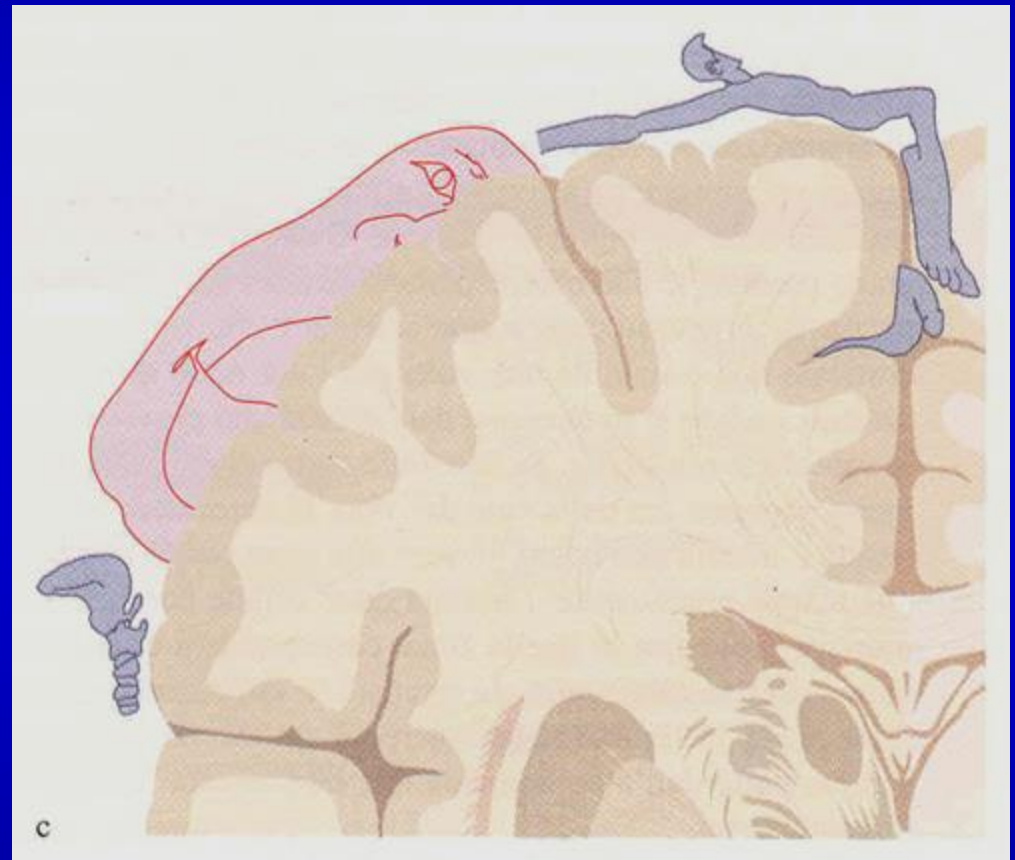
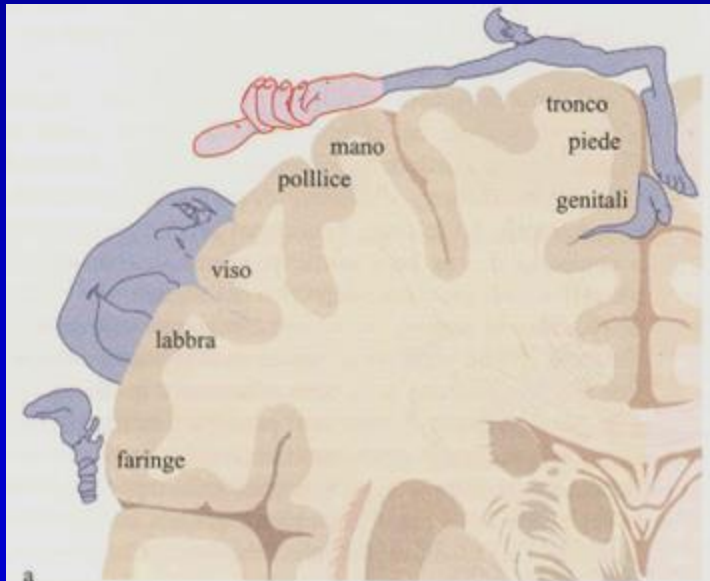
A Sensory homunculus



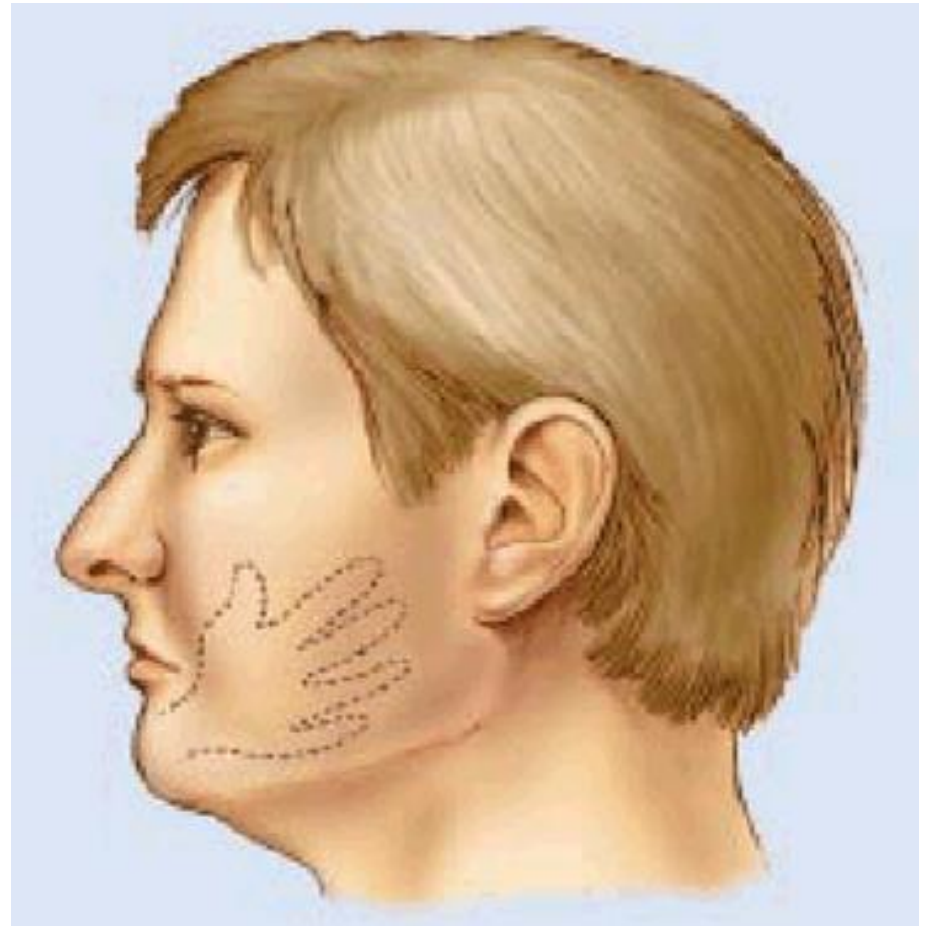
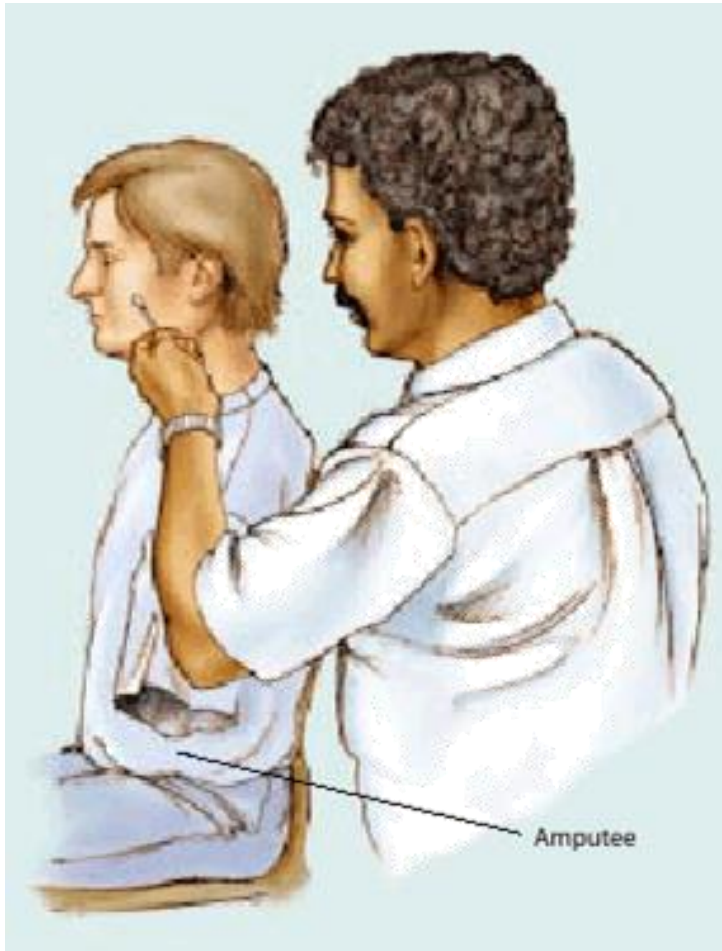
B Motor homunculus



Expansion of the functional map

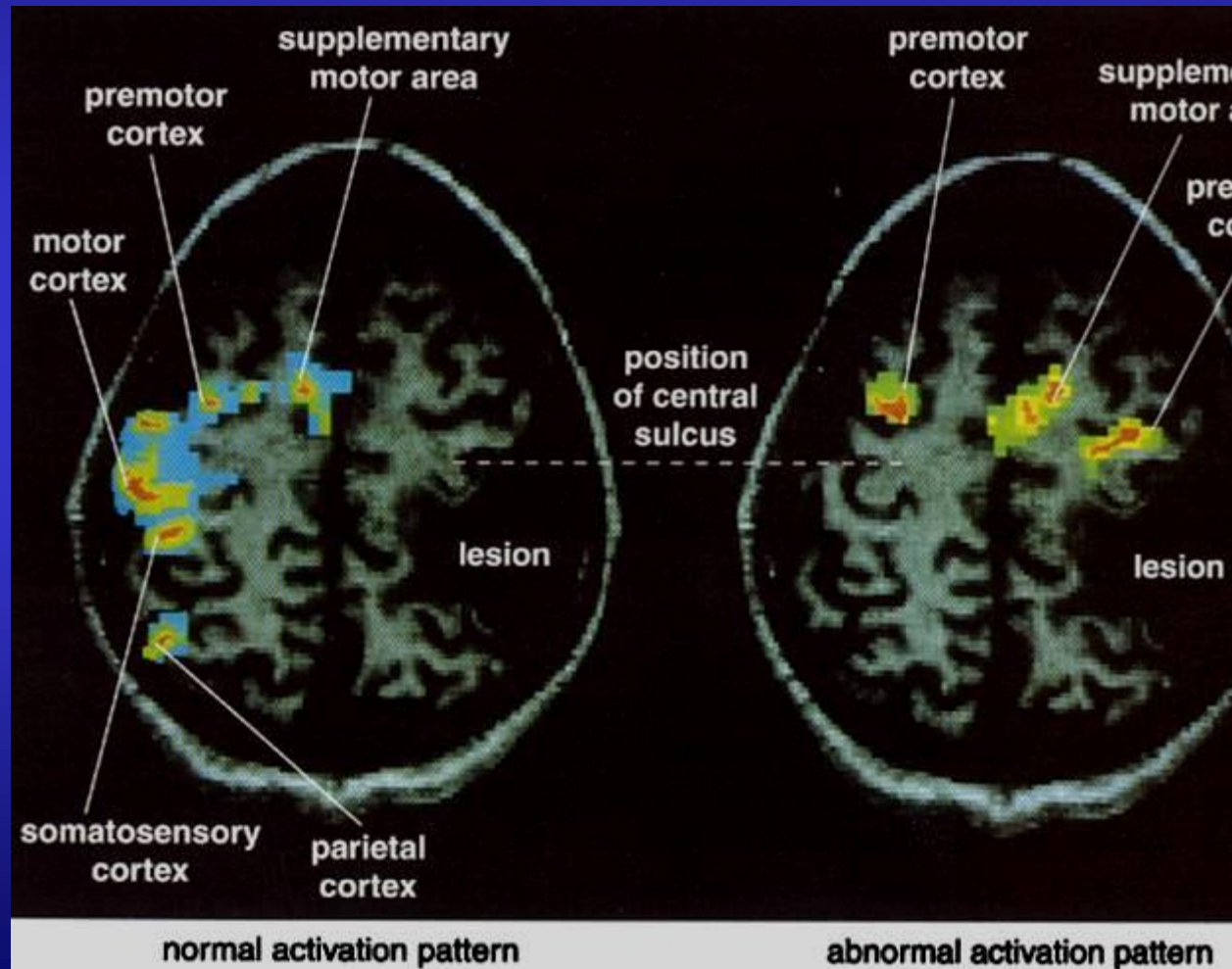


Ramachandran's procedure



Compensation?

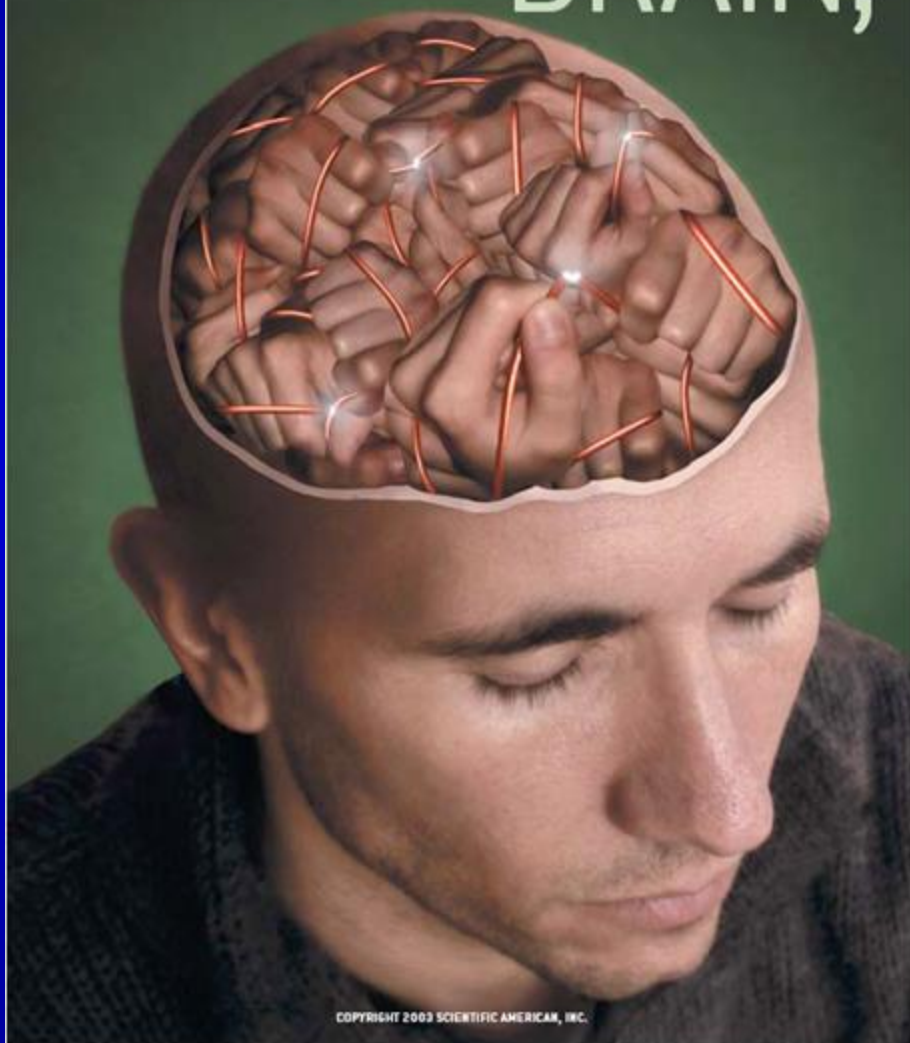
Abnormal neuronal activity in the premotor cortex and the supplementary motor area of the non-lesioned hemisphere suggests that these regions have learned to compensate for the damaged motor cortex



BRAIN,

REPAIR YOURSELF

HOW DO YOU FIX A BROKEN BRAIN? THE ANSWERS MAY LITERALLY LIE WITHIN OUR HEADS. THE SAME APPROACHES MIGHT ALSO BOOST THE POWER OF AN ALREADY HEALTHY BRAIN BY FRED H. GAGE



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NEUROSCIENCE

The human brain has the capability to rewire itself to some extent.

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FOR MOST OF ITS 100-YEAR HISTORY, NEUROSCIENCE has embraced a central dogma: a mature adult's brain remains a stable, unchanging, computerlike machine with fixed memory and processing power. You can lose brain cells, the story has gone, but you certainly cannot gain new ones. How could it be otherwise? If the brain were capable of structural change, how could we remember anything? For that matter, how could we maintain a constant self-identity?

Although the skin, liver, heart, kidneys, lungs and blood can all generate new cells to replace damaged ones, at least to a limited extent, until recently scientists thought that such regenerative capacity did not extend to the central nervous system, which consists of the brain and spinal cord. Accordingly, neurologists had only one counsel for patients: "Try not to damage your brain, because there is no way to fix it."

Within the past five years, however, neuroscientists have discovered that the brain does indeed change throughout life—and that such revision is a good thing. The new cells and connections that we and others have documented may provide the extra capacity the brain requires for the variety of challenges that individuals face throughout life. Such plasticity offers a possible mechanism through which the brain might be induced to repair itself after injury or disease. It might even open the prospect of enhancing an already healthy brain's power to think and ability to feel.

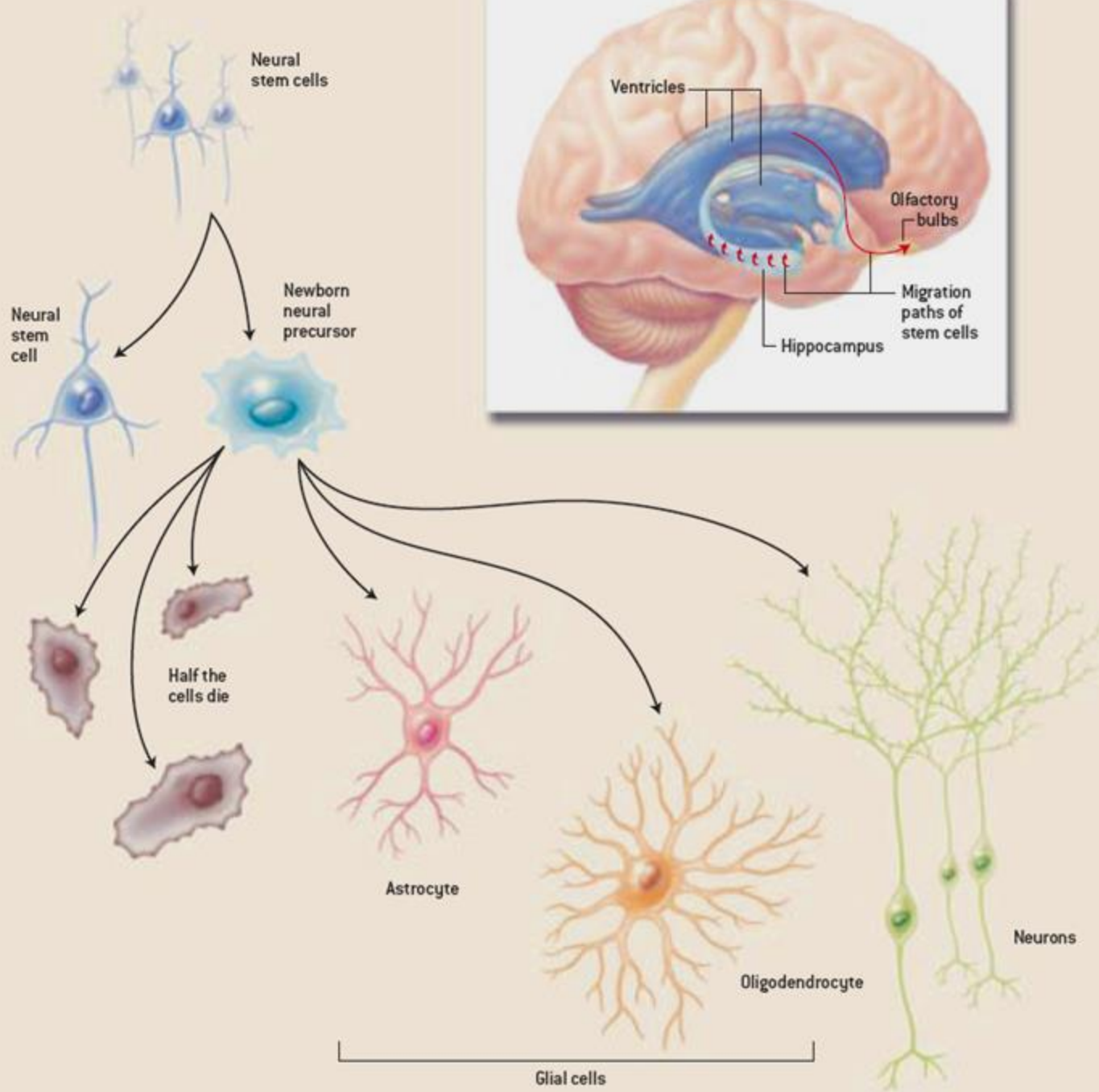
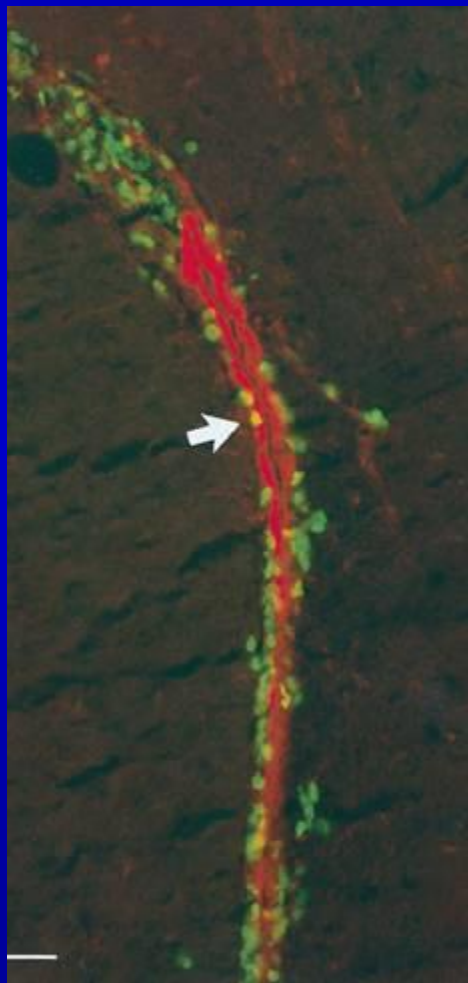
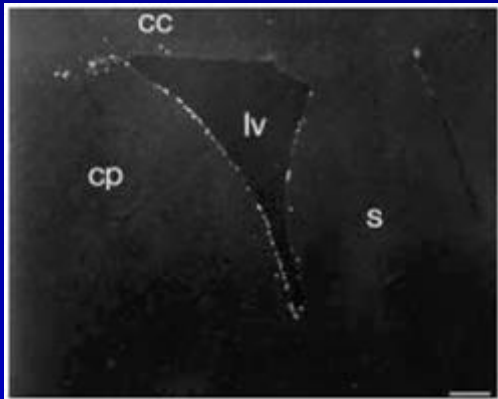
Neuroscientists, of course, have tried to come up with fixes for brain injury or brain disorders for decades. Such treatment strategies have primarily involved replacing diminished neurotransmitters, the chemicals that convey messages between nerve cells (neurons). In Parkinson's disease, for instance, a patient's brain loses the ability

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Neurogenesis in the adult brain

- Until very recently, a central dogma of neuroscience has been that new neurons are not added to the adult mammalian brain
- For more than 100 years it has been assumed that neurogenesis, or the production of new neurons, occurs only during development and stops before puberty
- Koelliker, His, Ramón y Cajal and others had described in detail the development of the central nervous system of humans and other mammals
- In the first half of the twentieth century, there were occasional reports of postnatal neurogenesis in mammals - the possibility that the new cells arising in the subependymal layer might migrate into the cerebrum to form mature neurons was raised



SELECTED NEURAL GROWTH FACTORS UNDER DEVELOPMENT

These factors might be used as drugs on their own, or scientists might design other drugs to stimulate or block the factors.

NAME	FUNCTION	POTENTIAL DISEASE TARGETS	SOME COMPANIES INVOLVED IN RESEARCH
Brain-derived neurotrophic factor (BDNF)	Keeps newborn neurons alive	Depression (abandoned for amyotrophic lateral sclerosis)	Amgen, Thousand Oaks, Calif.
Ciliary neurotrophic factor (CNTF)	Protects neurons from death	Huntington's disease (now testing against obesity)	Regeneron Pharmaceuticals, Tarrytown, N.Y.
Epidermal growth factor (EGF)	Spurs stem cells in brain to divide	Brain tumors and stroke	ImClone Systems, New York City
Fibroblast growth factor (FGF)	In low doses, supports survival of various cell types; at high doses, induces cells to proliferate	Brain tumors and stroke	ViaCell, Boston
Glial cell line–derived neurotrophic factor (GDNF)	Prompts motor neurons to sprout new branches; prevents cells that perish in Parkinson's disease from dying	Parkinson's disease and ALS	Amgen
Glial growth factor-2 (GGF-2)	Favors production of glial (support) cells	Spinal cord injury, multiple sclerosis and schizophrenia	Acorda Therapeutics, Hawthorne, N.Y.
Insulinlike growth factor (IGF)	Fosters the birth of both neurons and glial cells	Multiple sclerosis, spinal cord injury, ALS and age-related dementia	Cephalon, West Chester, Pa.
Neurotrophin-3 (NT-3)	Promotes formation of oligodendrocytes (type of glial cell)	Multiple sclerosis, spinal cord injury and ALS	Amgen and Regeneron Pharmaceuticals